

INTRODUCTION

Steefo Engineering Corporation is an engineering company with the State of the art manufacturing facility, specialized in the field of Rolling Mill & Billet Plant technology and is fully committed and dedicated to the steel and related Industries.

Supply of various equipment and process technology for Rolling Mill & Billet Plant are one of the core activities of Steefo Engineering and we design, manufacture and supply Turn Key and tailor-made systems for the Steel Plant.

In the last 45 years Steefo Engineering has acquired the technology and Know How for the entire production of quality steels by working with metallurgists and process experts who have worked and gained valuable experience in the metal production industry worldwide.

Steefo Engineering is able to design and built single units as well as whole production plants for the Rolling Mill & Billet Plant from one hand ensuring single cast design of the entire plant.

We have done Steel Plants of Reputed Groups on Turn Key basis in around 20 countries, including India, Indonesia, Saudi Arabia, Jordan, Qatar, Angola, D.R of Congo, Ethiopia, Mozambique, Kenya , Uganda; Zimbabwe, Mali, Nigeria, Ghana, Rwanda, Sri Lanka, Bangladesh, Gabon, Sudan, Benin, Libya, UAE and Mauritius.

All our above projects are working satisfactorily due to good quality of finish material, low cost of production, low maintenance and better after sales service.

GENERAL SPECIFICATIONS AND DESCRIPTION OF ROLLING MILL**1. TECHNICAL PARAMETERS**

Mill Production:	25 to 30 Tons / Hr.
Basic Raw Material:	Billets 100 & 130 mm
Finished Products:	8 mm Ø to 32mm High Strength TMT Re Bars Wire Rod Mill 5.5mm to 8mm

2. PRE-CONDITIONS OF BILLETS**TOLERANCES**

Square Billets:	Side length ± 1.5 % max. Diagonal < 6 mm
Cut to Length:	± 50 mm
Bending:	Not more than 10mm in 1m Not more than 40mm in 12m Billet without distinct twist, irrectangular Billet angular twist shall not be greater than $1^\circ/\text{m}$

3. STEEL GRADE	Common low and medium carbon steel for rebars according to standard BS 4449 Grade DIN 448
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4. TOLERANCES OF FINISHED PRODUCTS

Finished products according to normal B.S. standard or equivalent to any international standard.

5. STARTING TEMPERATURE

Billet temperature in Roughing Stand: $1,100^\circ\text{C}$ approx.



A. BASIC DATA AND MEDIA REQUIREMENTS**1. PLANT POWER CONDITION**

- 1.1 AC - High Voltage: 33 KV $\pm$ 10% - 3 Ph (or as per Local available)
- Low Voltage: 415 V, $\pm$ 10% 3 Ph
- Frequency: 50 Hz
1.2 - Solenoid Valves : 24V DC
- Limit Switches – Photocell: DC 24 V
- Control Circuit: DC 24V

2. CLEAN WATER SYSTEM (CLOSED CIRCUIT)

Water flow rate:
Mill Water : 200m³ / hr approx.
Operating pressure: approx. 4 bar for Mill
TMT Water : 500m³ / hr approx.
approx. 16 Bar for TMT
Temperature: max 35°C approx.

3. COMPRESSED AIR**Air Consumption**

- Free Air: 300 CFM

4. HYDRAULIC OIL

Initial filling oil for hydraulic 1000 litres (app.)
Central unit:
Enklo 68 (H.P.) / Equivalent

5. LUBRICATION OIL

Initial filling oil for lubrication
Unit Gear boxes etc. 4,000 Litres (app.)

Note: oil having the following properties will be used

- GEAR OIL VG - 32
- Pour point °C max: - 15
- Viscosity c at 40°C: 88.8
- Flash point °C: 246

6. LUBRICATION GREASE

Initial filing grease for central unit

- Lithon EP (IOC) or equivalent : 200 Kgs
- EP 350 (H.P.) : 100 Kgs
- High Temperature Moly Coat : 200 Kgs

7. ROLL CONSUMPTION: 0.4 to 0.6 Kg / Ton

8. POWER REQUIREMENT 3 MW



DESIGN PARAMETERS											
Finished Product Requirement		160,000	TPA			Per Hour Production	:	25 to 30		T/hr	
Billet Requirement		166,400	TPA			Maximim Finishing Speed	:	25		m/sec	
Billet Size		130	Sq.			Average Yield	:	96		%	
Billet Length		6	m								
Billet Weight		780	kg								
PRODUCTION TABLE - TMT BAR											
			Rolling	Rolling	Rolling	Th. Rolling	Actual Rolling	Annual Production		Annual	
Product		No. of Strand	Speed	Length	Time	Capacity	Capacity	Rolling	Billet Req.	Prod. Time	
Type	mm		kg	m/sec	m	sec	T/hr	T/hr	%age	TPA	Hrs.
TMT BARS	8	1	0.390	25	2000	80	35.10	25	15	166,400	960
	10	1	0.617	18	1264	70	39.98	30	20		1067
	12	1	0.888	12	878	73	38.36	30	20		1067
	16	1	1.578	7	494	71	39.77	32	20		1000
	20	1	2.466	4.5	316	70	39.95	32	15		750
	25	1	3.853	3	202	67	41.61	32	5		250
	32	1	6.313	2	124	62	45.45	32	5		250
								Total	100	166400	5343
Finished Production at an average 96% yield								:	160000		
No.of Rolling Hrs available in a year								:	7920		hrs
Hours required for production								:	5343		hrs
Mill Utilisation								:	67.47		%



AUTOMATIC BAR ROLLING MILL**SUMMARY****A. ROLLING MILL EQUIPMENT**

- | | |
|---|----------|
| 1. 550mm Heavy Duty Roughing Mill | 1 Stand |
| 2. 400mm Continuous Intermediate Mill | 2 Stands |
| 3. 360mm Continuous Intermediate Mill | 4 Stands |
| 4. 320mm Continuous Finishing Mill | 4 Stands |
| 5. 280mm Continuous Finishing Mill | 4 Stands |
| 6. Auxiliary Equipment | |
| 7. Electricals | |
| 8. Automatic Cooling Bed with Twin Channel & Cold Shear | |
| 9. Continuous Operating Dividing Shear for TMT Bars | |
| 10. Bar Finishing, Bundling & Tying Machine (Packing Bed) | |
| 11. TMT System | |

12. Central Oil Lubrication System

- | | |
|------------------------------|----------------|
| B. Wire Ro Block Mill | 6 Stand |
|------------------------------|----------------|

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| C. Consultation & Supervision of Erection, Installation & Commissioning of Rolling Mill | |
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A. Rolling Mill Equipment**1. 550mm Heavy Duty Roughing Mill****1.1 High Speed Flywheel****1 No**

Steel Flywheel running @ 740 rpm
app. 8 Tons weight complete with shaft,
spherical roller bearings, pedestals and base plate.

1.2 Reduction Gear Box.**2 Nos.**

Input - 740 rpm
Output - 95 rpm
Nominal Power - 1120 KW
Peak load - 2240 KW
Momentary Load - 3360 KW

Construction:

- Gears & Pinions will be of forged Steel, shrunk fitted on En-9 forged steel shafts.
- Helical teeth Profile on Gear Hobbing Machine
- Shafts end with key way.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Piping provided for forced lubrication arrangement of bearings and pinion teeth.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Pinion enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.

Transparent window provided on both sides of the housing to observe teeth and lubrication



1.3 550mm 3 -Hi Pinion Gear Box.

1 No.

Pinion Centre - 550 mm
Nominal Power - 1120 KW
Peak load - 2240 KW
Momentary Load - 3360 KW

Construction:

- Gears & Pinions will be of forged Steel, shrunk fitted on En-9 forged steel shafts.
- Helical teeth Profile on Gear Hobbing Machine
- Shafts end with key way.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Piping provided for forced lubrication arrangement of bearings and pinion teeth.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Pinion enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.

Transparent window provided on both sides of the housing to observe teeth and lubrication

1.4 Base Plate for Pinion Gear Box

2 Nos.



1.5 Mill Stand**1 No.**

- Type : 3 Hi
Size : 550MM
Roll Dia. : Max. 590 MM
Min. 530 MM
Roll Barrel Length : 1350 MM (or as per requirement)
Roll Neck Bearing : Spherical Roller Bearing
Housing & Cap : Cast Steel – Grade-2 (IS-1030)
All Fasteners : High Tensile Steel Gr.8.8 min.
Material for Chocks : Cast Steel with 500-600 N/mm² U.T.S.
Chocks are accurately machined and are inter changeable.
Top Roll Adjustment : By screw down mechanism.

- The housing shall be accurately machined on CNC machine and fitted with wear plates by high tensile Allen Cap Screw. This ensures accurate fitting of chocks in housings. The wear plates are ground Finished
- The cap shall be fixed over the housings by means of two taper wedges which gives very good rigidity and does not loose during Mill operation.
- Mill Stand will be complete with all fitments, with Cooling Water piping and valves, roll cooling and Guide Cooling and Foundation bolts etc.
- The accurate machining of parts and housings keep vibration level to minimum during rolling.

1.6 Cardan Shaft & Universal Head**3 Set**

- Location : Between Pinion Stand and rolls
Type : Cross Joint Type
Material of coupling heads : EN-9 (Forged)

CONSTRUCTION

- Cardan Shaft & Couplings are be required to connect output shafts of Pinion Stand to rolls of mill stand.
- Each set shall consist of two couplings hubs assemblies and one cardan Shaft.
- Cardan Shaft will be selected for the required duties from renowned manufacturers.
- The connecting Flange shall have keys to ensure positive transmission and reduce the loads on bolts.
- Connecting Hub suitable for roll end on one side & pinion end on the other side are provided.



- 1.7 Full Geared Heavy Duty Couplings** **4 Set**
Type : full flexible Geared Coupling to be connected between
Reduction Gear Box, Pinion Stand and Motor
- Design Parameters**
Normal Rating : 1120 KW
RPM : 750 RPM
Service Factor : ≥ 3
Type of Drive : Non reversible for all
Duty : Rolling Mill Duty
- 1.8 Foundation Bolts** **1 Set**
- 1.9 Motor Base Plate** **1 Set**



2. 400mm Continuous Intermediate Mill. (I-1 & I-2)**2.1 Reduction cum Pinion Gear Box****2 Nos.**

Combined Gear-Cum-2 High Pinion type designed according to DIN Standards.

Normal Rating	:	300 KW
RPM	:	750-1250
Service Factor	:	2.5 (On Load)
Duty	:	Rolling Mill
Bearings	:	SPH. Roller Bearings

Materials of construction are as follows:

- Gears & Pinions will be of forged 18CrNiMo-7/case Carburized steel hardened to 60±2 HRC, shrunk fitted on En-9 forged steel shafts. The gears and pinions will conform to DIN 3 / 4 Standard.
- Helical teeth hardened and Profile Ground Finish on Gear grinder.
- Shafts Heat Treated Steel.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Gears enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.
- Transparent window provided on both sides of the housing to observe teeth and lubrication.
- Bolts: High Tensile Steel
- Gearboxes will be complete with internal and external piping suitable for central oil lubrication at a supply pressure of 0.8 to 1.5 bar. An isolating valve, a Y-Strainers, Pressure Gauge, Pressure switch and a Flow Switch included in supply line.



2.2 400 mm Mill Stands**2 Nos.**

- Type : 2 Hi
Size : 400MM
Roll Dia. : Max. 440 MM
Min. 380 MM
Barrel : 800 MM (or as per requirement)
Roll Neck Bearing : Spherical Roller Bearing
Housing & Cap : Cast Steel – Grade-2 (IS-1030)
All Fasteners : High Tensile Steel Gr.8.8 min.
Material for Chocks : Cast Steel with 500-600 N/mm² U.T.S.
Chocks are accurately machined and are inter changeable.
Top Roll Adjustment : By screw down mechanism.
- The housing shall be accurately machined on CNC machine and fitted with wear plates by high tensile Allen Cap Screw. This ensures accurate fitting of chocks in housings. The wear plates are ground Finished
 - The cap shall be fixed over the housings by means of two taper wedges which gives very good rigidity and does not loose during Mill operation.
 - Mill Stand will be complete with all fitments, with Cooling Water piping and valves, roll cooling and Guide Cooling and Foundation bolts etc.
 - The accurate machining of parts and housings keep vibration level to minimum during rolling.

2.3 Cardan Shafts**4 Nos.**

- Location : Between Pinion Stand and rolls
Type : Cross Joint Type
Material of coupling heads : EN-9 (Forged)

CONSTRUCTION

- Cardan Shaft & Couplings are be required to connect output shafts of Pinion Stand to rolls of mill stand.
- Each set shall consist of two couplings hubs assemblies and one cardan Shaft.
- Cardan Shaft will be selected for the required duties from renowned manufacturers.
- The connecting Flange shall have keys to ensure positive transmission and reduce the loads on bolts.
- Connecting Hub suitable for roll end on one side & pinion end on the other side are provided.



2.4 Geared Coupling **2 Nos.**
Type : full flexible Geared Coupling to be connected between
Reduction Gear Box, Pinion Stand and Motor

Design Parameters

Normal Rating : 300 kW
RPM : 750 - 1250
Service Factor : ≥ 3
Type of Drive : Non reversible for all
Duty : Rolling Mill Duty

2.5 Foundation Bolts **1 Set**

2.6 Motor Base Plate **8 Nos.**



3. 360mm Continuous Intermediate Mill. (I-3, I-4, I-5 & I-6)**3.1 Reduction cum Pinion Gear Box****4 Nos.**

Combined Gear-Cum-2 High Pinion type designed according to DIN Standards.

Normal Rating	:	300 KW
RPM	:	750-1250
Service Factor	:	2.5 (On Load)
Duty	:	Rolling Mill
Bearings	:	SPH. Roller Bearings

Materials of construction are as follows:

- Gears & Pinions will be of forged 18CrNiMo-7/case Carburized steel hardened to 60±2 HRC, shrunk fitted on En-9 forged steel shafts. The gears and pinions will conform to DIN 3 / 4 Standard.
- Helical teeth hardened and Profile Ground Finish on Gear grinder.
- Shafts Heat Treated Steel.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Gears enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.
- Transparent window provided on both sides of the housing to observe teeth and lubrication.
- Bolts: High Tensile Steel
- Gearboxes will be complete with internal and external piping suitable for central oil lubrication at a supply pressure of 0.8 to 1.5 bar. An isolating valve, a Y-Strainers, Pressure Gauge, Pressure switch and a Flow Switch included in supply line.



3.2 360 mm Mill Stands**4 Nos.**

- Type : 2 Hi
Size : 360MM
Roll Dia. : Max. 400 MM
Min. 340 MM
Barrel : 700 MM (or as per requirement)
Roll Neck Bearing : Spherical Roller Bearing
Housing & Cap : Cast Steel – Grade-2 (IS-1030)
All Fasteners : High Tensile Steel Gr.8.8 min.
Material for Chocks : Cast Steel with 500-600 N/mm² U.T.S.
Chocks are accurately machined and are inter changeable.
Top Roll Adjustment : By screw down mechanism.
- The housing shall be accurately machined on CNC machine and fitted with wear plates by high tensile Allen Cap Screw. This ensures accurate fitting of chocks in housings. The wear plates are ground Finished
 - The cap shall be fixed over the housings by means of two taper wedges which gives very good rigidity and does not loose during Mill operation.
 - Mill Stand will be complete with all fitments, with Cooling Water piping and valves, roll cooling and Guide Cooling and Foundation bolts etc.
 - The accurate machining of parts and housings keep vibration level to minimum during rolling.

3.3 Cardan Shafts**8 Nos.**

- Location : Between Pinion Stand and rolls
Type : Cross Joint Type
Material of coupling heads : EN-9 (Forged)

CONSTRUCTION

- Cardan Shaft & Couplings are be required to connect output shafts of Pinion Stand to rolls of mill stand.
- Each set shall consist of two couplings hubs assemblies and one cardan Shaft.
- Cardan Shaft will be selected for the required duties from renowned manufacturers.
- The connecting Flange shall have keys to ensure positive transmission and reduce the loads on bolts.
- Connecting Hub suitable for roll end on one side & pinion end on the other side are provided.



3.4 Geared Coupling **4 Nos.**
Type : full flexible Geared Coupling to be connected between
Reduction Gear Box, Pinion Stand and Motor

Design Parameters

Normal Rating : 300 kW
RPM : 750 - 1250
Service Factor : ≥ 3
Type of Drive : Non reversible for all
Duty : Rolling Mill Duty

3.5 Foundation Bolts **1 Set**

3.6 Motor Base Plate **16 Nos.**



4. 320mm Continuous Intermediate Mill. (I-7, I-8, I-9 & I-10)**4.1 Reduction cum Pinion Gear Box****4 Nos.**

Combined Gear-Cum-2 High Pinion type designed according to DIN Standards.

Normal Rating	:	300 KW
RPM	:	1000-1800
Service Factor	:	2.5 (On Load)
Duty	:	Rolling Mill
Bearings	:	SPH. Roller Bearings

Materials of construction are as follows:

- Gears & Pinions will be of forged 18CrNiMo-7/case Carburized steel hardened to 60 ± 2 HRC, shrunk fitted on En-9 forged steel shafts. The gears and pinions will conform to DIN 3 / 4 Standard.
- Helical teeth hardened and Profile Ground Finish on Gear grinder.
- Shafts Heat Treated Steel.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Gears enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.
- Transparent window provided on both sides of the housing to observe teeth and lubrication.
- Bolts: High Tensile Steel
- Gearboxes will be complete with internal and external piping suitable for central oil lubrication at a supply pressure of 0.8 to 1.5 bar. An isolating valve, a Y-Strainers, Pressure Gauge, Pressure switch and a Flow Switch included in supply line.



4.2 320 mm Mill Stands**4 Nos.**

- Type : 2 Hi
Size : 320MM
Roll Dia. : Max. 360 MM
Min. 300 MM
Barrel : 600 MM (or as per requirement)
Roll Neck Bearing : Spherical Roller Bearing
Housing & Cap : Cast Steel – Grade-2 (IS-1030)
All Fasteners : High Tensile Steel Gr.8.8 min.
Material for Chocks : Cast Steel with 500-600 N/mm² U.T.S.
Chocks are accurately machined and are inter changeable.
Top Roll Adjustment : By screw down mechanism.
- The housing shall be accurately machined on CNC machine and fitted with wear plates by high tensile Allen Cap Screw. This ensures accurate fitting of chocks in housings. The wear plates are ground Finished
 - The cap shall be fixed over the housings by means of two taper wedges which gives very good rigidity and does not loose during Mill operation.
 - Mill Stand will be complete with all fitments, with Cooling Water piping and valves, roll cooling and Guide Cooling and Foundation bolts etc.
 - The accurate machining of parts and housings keep vibration level to minimum during rolling.

4.3 Cardan Shafts**8 Nos.**

- Location : Between Pinion Stand and rolls
Type : Cross Joint Type
Material of coupling heads : EN-9 (Forged)

CONSTRUCTION

- Cardan Shaft & Couplings are be required to connect output shafts of Pinion Stand to rolls of mill stand.
- Each set shall consist of two couplings hubs assemblies and one cardan Shaft.
- Cardan Shaft will be selected for the required duties from renowned manufacturers.
- The connecting Flange shall have keys to ensure positive transmission and reduce the loads on bolts.
- Connecting Hub suitable for roll end on one side & pinion end on the other side are provided.



4.4 Geared Coupling **4 Nos.**
Type : full flexible Geared Coupling to be connected between
Reduction Gear Box, Pinion Stand and Motor

Design Parameters

Normal Rating : 300 kW
RPM : 1000 - 1800
Service Factor : ≥ 3
Type of Drive : Non reversible for all
Duty : Rolling Mill Duty

4.5 Foundation Bolts **1 Set**

4.6 Motor Base Plate **16 Nos.**



5. 280mm Continuous Finishing Mill. (F-11, F-12, F-13 & F14)**5.1 Reduction cum Pinion Gear Box****4 Nos.**

Combined Gear-Cum-2 High Pinion type designed according to DIN Standards.

Normal Rating	:	300 KW
RPM	:	1000-1800
Service Factor	:	2.5 (On Load)
Duty	:	Rolling Mill
Bearings	:	SPH. Roller Bearings

Materials of construction are as follows:

- Gears & Pinions will be of forged 18CrNiMo-7/case Carburized steel hardened to 60±2 HRC, shrunk fitted on En-9 forged steel shafts. The gears and pinions will conform to DIN 3 / 4 Standard.
- Helical teeth hardened and Profile Ground Finish on Gear grinder.
- Shafts Heat Treated Steel.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Gears enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.
- Transparent window provided on both sides of the housing to observe teeth and lubrication.
- Bolts: High Tensile Steel
- Gearboxes will be complete with internal and external piping suitable for central oil lubrication at a supply pressure of 0.8 to 1.5 bar. An isolating valve, a Y-Strainers, Pressure Gauge, Pressure switch and a Flow Switch included in supply line.



5.2 280mm Mill Stands**4 Nos.**

- Type : 2 Hi
Size : 280MM
Roll Dia. : Max. 310 MM
Min. 260 MM
Barrel : 500 MM (or as per requirement)
Roll Neck Bearing : Roller Bearing
Housing & Cap : Cast Steel – Grade-2 (IS-1030)
All Fasteners : High Tensile Steel Gr.8.8 min.
Material for Chocks : Cast Steel with 500-600 N/mm² U.T.S.
Chocks are accurately machined and are inter changeable.
Top Roll Adjustment : By screw down mechanism.
- The housing shall be accurately machined on CNC machine and fitted with wear plates by high tensile Allen Cap Screw. This ensures accurate fitting of chocks in housings. The wear plates are ground Finished
 - The cap shall be fixed over the housings by means of two taper wedges which gives very good rigidity and does not loose during Mill operation.
 - Mill Stand will be complete with all fitments, with Cooling Water piping and valves, roll cooling and Guide Cooling and Foundation bolts etc.
 - The accurate machining of parts and housings keep vibration level to minimum during rolling.

5.3 Cardan Shafts**8 Nos.**

- Location : Between Pinion Stand and rolls
Type : Cross Joint Type
Material of coupling heads : EN-9 (Forged)

CONSTRUCTION

- Cardan Shaft & Couplings are be required to connect output shafts of Pinion Stand to rolls of mill stand.
- Each set shall consist of two couplings hubs assemblies and one cardan Shaft.
- Cardan Shaft will be selected for the required duties from renowned manufacturers.
- The connecting Flange shall have keys to ensure positive transmission and reduce the loads on bolts.
- Connecting Hub suitable for roll end on one side & pinion end on the other side are provided.



5.4 Geared Coupling **4 Nos.**
Type : full flexible Geared Coupling to be connected between
Reduction Gear Box, Pinion Stand and Motor

Design Parameters

Normal Rating : 300 kW
RPM : 1000 - 1800
Service Factor : ≥ 3
Type of Drive : Non reversible for all
Duty : Rolling Mill Duty

5.5 Foundation Bolts **1 Set**

5.6 Motor Base Plate **16 Nos.**



6. Auxiliary Equipment**6.1 Y Table for Roughing Mill to feed Top passes automatic**

Purpose : To receive the 1st & 3rd Pass and convey the material to Y troughs to feed to top passes of mill stands

Quantity : One (1) Set

Length : 10000 mm Appx.

Roller Spacing : 1000mm

Rollers

- Diameter : 268mm

- Barrel Length : 1200 mm

No. of Rollers : Nine (9) Nos.

Speed : 3.0 m/sec

Shaft Bearing : Spherical Roller Bearings

Frame : Fabricated from steel structure

Drive : Group driven with chain and sprockets through worm reducer

Motor kw : 11 KW

Motor RPM : 1000 RPM

Material of construction

Frame : Mild Steel Construction of E 250 as per IS : 2062

Sprocket : 45C8 as per IS:1570 Induction hardened to 250-280 BHN

Shaft : 40C8 as per IS : 1570

Rollers : Seamless Pipes



6.2 Fix Roller Table for Roughing Mill to feed Bottom passes automatic

Purpose	:	To Feed Bottom Passed on Roughing Mill
Quantity	:	One (1) Set
Length	:	8000 mm Appx.
Roller Spacing	:	1000mm
Rollers		
- Diameter	:	268mm
- Barrel Length	:	1200 mm
No. of Rollers	:	Nine (9) Nos.
Speed	:	3.0 m/sec
Shaft Bearing	:	Spherical Roller Bearings
Frame	:	Fabricated from steel structure
Drive	:	Group driven with chain and sprockets through worm reducer
Motor kw	:	11 KW
Motor RPM	:	1000 RPM

Material of construction

Frame	:	Mild Steel Construction of E 250 as per IS : 2062
Sprocket	:	45C8 as per IS:1570 Induction hardened to 250-280 BHN
Shaft	:	40C8 as per IS : 1570
Rollers	:	Seamless Pipes
Apron Plate	:	Minimum 20 mm Fabricated from Mild Steel Plates



6.3 Approach Roller Table to 1st Continuous Mill (Extension Roller Table)

Purpose	:	To handle the material after roller table with flappers
Total Length	:	15000 mm Appx.
Table Length (C/C to end rollers)	:	15,000 mm
Roller Spacing	:	1000 mm
Rollers		
- Diameter	:	246 mm
- Barrel Length	:	700 mm
No. of Rollers	:	Sixteen (16) Nos.
Shaft Bearing	:	Spherical Roller Bearings - 22220
Roller Drive	:	Group driven with Chain & Sprocket through suspended gearbox
Motor KW	:	11 KW
Overall reduction ratio	:	4.5
Motor RPM	:	1000 RPM
Peripheral Speed of Rollers	:	3.0 m/s
Material of construction		
Frame	:	Mild Steel Construction of E 250 as per IS : 2062
Sprocket	:	45C8 as per IS:1570 Induction hardened to 250-280 BHN
Shaft	:	40C8 as per IS: 1570
Rollers	:	Seamless Pipes
Apron Plate	:	Minimum 20mm Fabricated from Mild Steel Plates



6.4 Pinch roll Before Quenching Box

Purpose	:	To pinch and convey the high-speed bars
Location	:	In between TMT Line
Quantity	:	One (1) No.
Type	:	Cantilever
Pinch Roll Dia	:	Max. Ø 310 mm Min. Ø 270 mm
Barrel Length	:	Pinch Roll - 70mm
Bar Speed	:	12 m/sec-22 m/sec
Motor	:	89 KW
RPM	:	1500 RPM
Drive	:	With reduction gearbox and cardan shaft (top roll pneumatically operated)
Ratio	:	As per requirement
Pinching	:	Pneumatic by operation of Top Roll
Lubrication	:	Splash Oil/Forced
Bearings	:	SKF/FAG Make
Body Construction	:	Structural Steel Fabricated, duly stress relieved & machined
Common Base Frame for Pinch Roll & Motor	:	Structural Steel Fabricated, duly stress relieved & machined
Drive Coupling	:	Flexible Gear Coupling
Gears Material	:	18CrNiMo7-6 case hardened & heat treated
Shafts Material	:	18CrNiMo7-6 case hardened & heat treated
Entry & Exit Guides	:	Cast Steel/Steel Fabricated

Complete with pneumatic equipment comprising FRL unit, directional valves, pipes, hoses & fittings



6.5 Pinch roll after Quenching Box

Purpose	:	To pinch and convey the high-speed bars
Location	:	In between TMT Line
Quantity	:	One (1) No.
Type	:	Cantilever
Pinch Roll Dia	:	Max. Ø 310 mm
Min. Ø 270 mm		
Barrel Length	:	Pinch Roll - 70mm
Bar Speed	:	12 m/sec-22 m/sec
Motor	:	89 KW
RPM	:	1500 RPM
Drive	:	With reduction gearbox and cardan shaft (top roll pneumatically operated)
Ratio	:	As per requirement
Pinching	:	Pneumatic by operation of Top Roll
Lubrication	:	Splash Oil/Forced
Bearings	:	SKF/FAG Make
Body Construction	:	Structural Steel Fabricated, duly stress relieved & machined
Common Base Frame for Pinch Roll & Motor	:	Structural Steel Fabricated, duly stress relieved & machined
Drive Coupling	:	Flexible Gear Coupling
Gears Material	:	18CrNiMo7-6 case hardened & heat treated
Shafts Material	:	18CrNiMo7-6 case hardened & heat treated
Entry & Exit Guides	:	Cast Steel/Steel Fabricated

Complete with pneumatic equipment comprising FRL unit, directional valves, pipes, hoses & fittings



6.6 Start/ Stop type Crop & Cobble Shear – 2 Nos.

For Hot Material Cutting (Temp. 850°C to 1050°C)

Type : Direct Driven Dividing Shear (Steel Fabricated)

Motor : DC Motor with drive & Base Plate

Low Speed Start / Stop Shear with Lubrication Piping, Oil Supply System,
Delivery Guide Trough

Gears : Case Hardened & Ground with Surface Hardness HRC 58-60

Shafts : Heat Treated Steel

Blades : Heat Treated Steel (46-50 HRC)

Maximum Cross Sectional Area of the bar to : 3200 mm sq to be cut

Design Shear Strength of Bar at operating : 125 N/mm² temperature

Design Cutting Force : 400KN

Bar Speed : 0.8 - 1.3m/s

Each Shear complete with:

- Lubricating Piping on machine suitable for connection to central oil supply system.
- Entry & Exit Guides
- Crop Collection System including Steel fabricated chutes & buckets, pneumatic drop Guide



6.7 Tail Breaking Pinch Rolls**– 2 Nos.**

Purpose	: To pinch and feed the material to Twin channels
Location	: At entry of twin channels
Type	: Cantilever
Pinch Roll Diameter	: Max. Ø 300mm Min. Ø270mm
Barrel Length	: 70 mm
Bar Speed	: 12m/sec -22m/sec
Motor Power	: 89 KW
Motor Speed	: 1500 RPM
Drive	: With reduction gearbox and cardon shaft (top roll pneumatically operated)
Ratio	: As per requirement
Pinching	: Pneumatic by operation of Top Roll
Shifting of Pinch Rolls	: Hydraulic by operation
Lubrication	: Forced oil lubrication from Central Oil Lubrication System # 3.
Bearings	: Structural, SKF/FAG Make
Body Construction	: Structural Steel Fabricated, duly stress relieved & machined
Common Base Frame for Pinch Roll & Motor	: Structural Steel Fabricated, duly stress relieved & machined
Drive Coupling	: Flexible Gear Coupling
Gears Material	: 18CrNiMo7-6 case hardened & heat treated
Shafts Material	: 18CrNiMo7-6 case hardened & heat treated
Entry & Exit Guides	: Cast Steel/Steel Fabricated

Complete with pneumatic equipment comprising FRL unit, directional valves, pipes, hoses & fittings

6.8 Roller Guide Box with Holder**1 Set****6.9 Roller Twist Guide****1 Set****6.10 Funnels****1 Set****6.11 Static Guides****1 Set****6.12 Water pump for Rolling Mill
(1 Working + 1 Stand by)****2 Nos.**

7. ELECTRICALS

1.	AC Motor 1120 KW for 550mm Roughing Mill	1 No.
2.	300 KW D.C. Motor for 400mm Continuous Mill	2 Nos.
3.	300 KW D.C. Motor for 360mm Continuous Mill	4 Nos.
4.	300 KW D.C. Motor for 320mm Continuous Mill	4 Nos.
5.	300 KW D.C. Motor for 280mm Continuous Mill	4 Nos.
6.	Small AC Motors for Y Table, Turning Wall Roller table, Approach Roller table, etc.	1 Set
7.	Main LT A.C. Incomer Panel for all AC / DC Motors	1 Set
8.	D.C Drive Control Panel with Control Desk for Shears & Pinch Rolls	7 Nos.
9.	Scada base DC Drives Control Panel with Control Desk for 400mm, 360mm, 320mm & 280mm Continuous Mill	14 Nos.
10.	PLC, PCs, Remote I/Os etc System for Rolling Mill	1 Set
11.	Motorized Liquid Resistance starter for 1120 KW Motor	1 No.
12.	DC Motors for Shears & Pinch Roll	7 Nos.
13.	ACB for DC Incomer	1 No.
14.	APFC Panel with Capacitors	1 No.

Note : MV Side, Transformer, all Cables & Earthing's are not included



PRELIMINARY AC & DC MOTOR LIST										
MACHINE										
Item No.	Designation	Qty	Rated Each (KW)	Total Connected (KW)	Rated Speed (RPM)	Simultaneous working (KW)	Type of Motor	Voltage (V)	Type of Driven/ Duty	LOCATION
1.00	ROUGHING MILL									
1.01	550 MM 3-Hi Mill Stand	1	1120	1120	750	1120	DC	400	THY-2Q	DRIVE PANEL
1.02	Crop-Cum Cobble Shear -1	1	110	110	0/600	110	DC	400	THY-4Q	DRIVE PANEL
1.03	Lubrication Unit	1	7.5	15	1440	30	IM	415	S / D	MCC#1
2.00	INTERMEDIATE MILL 400mm									
2.01	400 MM 2-Hi Mill Stands	2	300	600	0/750/1250	2100	DC	400	THY-2Q	DRIVE PANEL
2.02	Lubrication Unit	1+1	7.5	15	1440	15	IM	415	S / D	MCC#1
2.03	Crop-Cum Cobble Shear	1	130	130	0/600	130	DC	400	THY-4Q	DRIVE PANEL
3.00	INTERMEDIATE MILL 360mm									
3.01	360 MM 2-Hi Mill Stands	4	300	1200	0/750/1250	2100	DC	400	THY-2Q	DRIVE PANEL
3.02	Lubrication Unit	1+1	7.5	15	1440	15	IM	415	S / D	MCC#1
3.03	Crop-Cum Cobble Shear -2	1	130	130	0/600	130	DC	400	THY-4Q	DRIVE PANEL
4.00	INTERMEDIATE MILL 320mm									
4.01	320 MM 2-Hi Mill Stand	4	300	1200	0/1000/1800	2100	DC	400	THY-2Q	DRIVE PANEL
4.02	Lubrication Unit	1+1	7.5	15	1440	15	IM	415	S / D	MCC#1
5.00	FINISHING MILL 280mm									
5.01	320 MM 2-Hi Mill Stand	4	300	1200	0/1000/1800	2700	DC	400	THY-2Q	DRIVE PANEL
5.02	Lubrication Unit	1+1	7.5	15	1440	15	IM	415	S / D	MCC#1



PRELIMINARY AC & DC MOTOR LIST										
MACHINE										
Item No.	Designation	Qty	Rated Each (KW)	Total Connected (KW)	Rated Speed (RPM)	Simultaneous working (KW)	Type of Motor	Voltage (V)	Type of Driven/Duty	LOCATION
6	STRAIGHT TMT LINE									
6.01	Pre & Post Pinch Roll	2	45	90	0/1800	135	DC	400	THY-2Q	DRIVE PANEL
6.02	Flying Shear Crank Type	1	270	270	0/1000	270	DC	400	THY-2Q	DRIVE PANEL
6.03	Lub Unit For Shears & Pinch Roll	1+1	5.5	5.5	1440	5.5	IM	415	DOL	DRIVE PANEL
6.04	Tail Brake Pinch Roll	2	60	120	0/1800	240	DC	400	THY-2Q	DRIVE PANEL
6.05	Twin Channel	1	15	15	1440	36	SC	415	DOL	DRIVE PANEL
7	COOLING BED (66 M)									
7.01	Aligning Rollers	28	1.1	30.8	110	30.8	GM	415	VVFD	DRIVE PANEL
7.02	Cooling Bed Rake Drive	1	73	73	0/1000	80	DC	400	THY-4Q	DRIVE PANEL
7.03	Layer Former (Fixed Type)	2	7.5	15	21	15	GM	415	VVFD	DRIVE PANEL
7.04	Bar transfer Arrangement in Layers	2	22.5	45	47	45	GM	415	VVFD	DRIVE PANEL
7.05	Run Out Roller Table	58	2.2	127.6	150	127.6	GM	415	VVFD	DRIVE PANEL
8	COLD SHEAR AREA									
8.01	Cold Shear	1	30	30	1440	60	IM	415	S / D	MCC#2
8.02	Lubrication Unit for Cold Shear	1	1.5	1.5	1440	1.5	IM	415	DOL	MCC#2
8.03	Cold Shear Run Out Roller Table (Depressing Table) – 6 Mtr	5	2.2	11	150	11	GM	415	VVFD	DRIVE PANEL
9	MATERIAL HANDLING AREA									
9.01	Chain Transfer Run In Roller Table	21	2.2	46.2	150	46.2	GM	415	VVFD	DRIVE PANEL
9.02	Chain Transfer No.-1	2	11	22	75	22	GM	415	VVFD	DRIVE PANEL
9.03	Chain Transfer No.-2	2	7.5	15	75	15	GM	415	VVFD	DRIVE PANEL
9.04	Bundle Forming Devices	1	7.5	7.5	75	7.5	SC	380	VVFD	DRIVE PANEL
9.05	Roller Table below Bundle Former	11	2.2	24.2	150	24.2	GM	415	VVFD	DRIVE PANEL
9.06	Hyd. Power Pack for Bar Handling Area	1+1	37	74	1440	74	IM	415	S/D	MCC#2
9.07	Tying M/c	1	13KVA			0			FEEDER	DRIVE PANEL
9.08	Tying Machine Run In Roller Table	11	2.2	24.2	150	24.2	GM	415	VVFD	DRIVE PANEL
9.09	Tied bundle Collection Roller Table	11	2.2	24.2	150	24.2	GM	415	VVFD	DRIVE PANEL
9.10	Bundle Storage Device	2	15	30	960	30	IM	415	VVFD	DRIVE PANEL



AC MOTOR TECHNICAL DETAIL 1120 KW		
SL.	PARTICULARS	
1	Type	WOUND
2	Mounting (IM)	1001
3	Index of Protection (IP)	23
4	Index of Cooling (IC)	01
5	Rating (kW)	1125
6	Polarity	8
7	Frame	KCW560C3
8	Rated Voltage(V)	415 +/- 10%
9	Frequency(Hz)	50 +/- 5%
10	Rated current (A)	1992
11	Rated Speed (RPM)	741
12	Rotor Voltage (V)	1120
13	Rotor Current (A)	620
14	Pull Out Torque (P.U)	2.20
15	Percentage Efficiency	
	Full load	95.8
	$\frac{3}{4}$ FL	95.5
	$\frac{1}{2}$ FL	94.0
16	Power Factor	
	Full Load	0.82
	$\frac{3}{4}$ FL	0.78
	$\frac{1}{2}$ FL	0.70
17	Insulation Class Stator/ Rotor	F/F
18	Temperature rise Rotor/ Stator(°C)	95/95
19	Method of Starting	LRS
20	Duty	S1
21	Amb Temperature (°C)	50
22	Noise & Vibration Limit	AS PER IS 325
23	Drive transmission	DIRECT FLEXIBLE



Sl No	Description	Qty
1	<u>DC Variable speed drive for 350 Kw , 450 VDC, 800 Amp DC motor for Rolling Stand</u> All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec . Field Supply is integrated in the Drive module. Drives are of CLASS IV overload capacity.	14 Set
A	- Parker DC Variable Speed , 599 series- Digital Microprocessor based Thyristor DC Drive of 1050 Amp with Class 4 Overload , suitable for 350 KW, 2Q DC Motor for stands. The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 630 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor. - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	14 Sets

Sl No	Description	Qty
2	<u>DC Variable speed drive for 80 Kw , 440 VDC, 200 Amp DC motor for CC Shear-1</u> All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec . Field Supply is integrated in the Drive module. Drives are of CLASS IV overload capacity.	1 Set
a	- Parker DC Variable Speed , 598 series- Digital Microprocessor based Thyristor DC Drive of 400 Amp with Class 4 Overload , suitable for 80 KW, 2Q DC Motor for CC Shear. The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 1250 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection.- - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor.- - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	1 Set

Sl No	Description	Qty
3	<u>DC Variable speed drive for 130 Kw , 440 VDC, 200 Amp DC motor for CC Shear-2</u> All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec . Field Supply is integrated in the Drive module. Drives are of CLASS IV overload capacity.	1 Set
a	- Parker DC Variable Speed , 598 series- Digital Microprocessor based Thyristor DC Drive of 400 Amp with Class 4 Overload , suitable for 80 KW, 2Q DC Motor for CC Shear. The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 1250 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection.- - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor.- - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	1 Set

4. DC Variable speed drive for 89 Kw , 440 VDC, 190 Amp DC motor for Pinch roll

All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec . Field Supply is integrated in the Drive module
Drives are of CLASS IV overload capacity.

A	- Parker DC Variable Speed , 591 series- Digital Microprocessor based Thyristor DC Drive of 270 Amp Continuous Current, suitable for 89 KW, 2QDC Motor for Pinch roll . The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 160 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor.- - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	2 Sets
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4. DC Variable speed drive for 89 Kw , 440 VDC, 190 Amp DC motor for TB**- 2 Set**

All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec . Field Supply is integrated in the Drive module.

Drives are of CLASS IV overload capacity

a	- Parker DC Variable Speed , 590 series- Digital Microprocessor based Thyristor DC Drive of 270 Amp Continuous Current, suitable for 89 KW, 4QDC Motor for TB . The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 200 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor. - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	2 Sets
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2) DC Variable speed drive for 225 Kw , 440 VDC, 600 Amp DC motor for TMT Shear 1 Set

All the rives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec. Field Supply is integrated in the Drive module. Drives are of CLASS IV overload capacity

A	- Parker DC Variable Speed , 598 series- Digital Microprocessor based Thyristor DC Drive of 1250 Amp With Class 4 Overload , suitable for 225 KW, 2QDC Motor for TB . The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 800 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor. - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	1 Sets
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6 DC Variable speed drive for 55 Kw , 440 VDC, 130 Amp DC motor

- 1 Set

For Cont Shear All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec . Field Supply is integrated in the Drive module. Drives are of CLASS IV overload capacity.

A	- Parker DC Variable Speed , 590 series- Digital Microprocessor based Thyristor DC Drive of 270 Amp Continuous Current, suitable for 55 KW, 4QDC Motor for Continuous shear. The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 200 Amp MCCB for control and Isolation of each DC Drive - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor. - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars.	1 Sets
B	- Switchgear of Siemens/Schneider/L & T Make - Mitsubishi servo drive for shifter - Mitsubishi , Servo Motor - Gearhead - With Motor power and resolver feedback cable	

7. DC Variable speed drive for 15 Kw , 440 VDC, 40 Amp DC motor for Twin Channel

1-Set

All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec. Field Supply is integrated in the Drive module Drives are of CLASS IV overload capacity.

A	- Parker DC Variable Speed , 590 series- Digital Microprocessor based Thyristor DC Drive of 110 Amp Continuous Current, suitable for 15 KW, 4QDC Motor for Twin Channel . The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 50 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor. - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	1 Sets
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8. DC Variable speed drive for 55 Kw , 440 VDC, 130 Amp DC motor for Rake

- 1 Set

All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec. Field Supply is integrated in the Drive module. Drives are of CLASS IV overload capacity.

A	- Parker DC Variable Speed , 590 series- Digital Microprocessor based Thyristor DC Drive of 270 Amp Continuous Current, suitable for 55 KW, 4QDC Motor for Rake . The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 200 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor. - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	1 Sets
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Sl No	Description	Qty
9	Common Incomer	2 Set
a	6300 Amp , EDO type ACB as common incomer	2 Set
b	Energy meter etc on the panel door	2 set
10	Control Desk	1 Set
a	Fabricated Control desk with Push buttons and Meters for the control of Stands, CC shear ,	
b	Siemens Win cc Scada	
c	Siemens 300 Series PLC	
11	Loose Supply Items	1 Set
a	Encoder with Coupling	28 Set
b	Profibus Cable	1 Lot
c	HMD	3 Nos.
d	Proximity Switch	6 No
12	Loop Scanner 2 Nos	1 Set
13	Essential Commissioning Spares	1 Set
a	Control card	1 Set
b	Power card for stack controller	1 No
c	Calibration card for stack controller	1 No
d	Encoder card	1 No
e	Power card for 270 Amp 4 Q Dc drive	1 No
f	Thyristor for 350 Kw stand drive	2 Nos
g	Thyristor for 270 Amp drive	2 Nos
h	Fuse for the stack of stand drive (1000 Amp)	1 No
i	Fuse for the field supply (40 Amp)	2 Nos



8. Automatic Cooling Bed with Twin Channel & Cold Shear
Size 72 Mtr.

a. Twin Channel

Application : Delivery of Rolled and cut Bars on to the Cooling Bed
Type : Electric Motor Driven single shaft Twin Channel
Bar Size : Dia 8mm – 32mm
Length : 72 Mtrs.
Maximum Speed : 25 M/Sec
Channels : S.G Iron Casting 1.1m long each
Central Box : 2.2m long Cast Iron, Air Cooled
Shafts : 2.2m Meter Long steel shaft
Bearings : Supported by pillow block
Cam : Forged Steel
Cam Followers: Roller Bearing Mounted on Pins in the C Channel
Coupling : Flexible
Drive : DC Electric Motor for line shaft operation

The Twin Channel bar delivery equipment consists of the following

- C type S.G. Iron Channels on both sides butting against cast Iron Central Box and hinged on top of the Box
- Box suspended through lugs from longitudinal frame on top
- Cam follower rollers mounted on a lug on each channel
- Single shaft is mounted on bearings and suspended from the longitudinal frame
- Cams for each C Channel are mounted on shaft
- Electric motor drive to main single shaft for movement

Operation

Both the C Channel (Right & Left) are closed. Rolled Re bar enters one of the C Channel and is cut to Cooling bed length while travelling by dividing shear. After cut, the bar tail end is braked using the tail braking pinch roll, so that the bar stops before its front end can shoot out of the channel. Motor operated cam shaft opens the C Channel to drop the bar on the Cooling bed straightening pockets. After dropping the bar, the channel closes immediately through rotation and is ready for the next bar. As soon the bar is cut, the next part of the bar is directed toward the other channel, which is ready to receive the same.



b. AUTOMATIC RAKE TYPE COOLING BED

Type	: Rake Type
Size	: 72 m long
Bar speed	: 25 m/sec (max)
In-let temperature of bar	: 600°C
Out-let temperature of bar	: 250°C

The cooling bed proposed is of automatic rake type and has the following:-

- i) Fixed Grid
- ii) Moving rakes
- iii) Eccentrics and drives.
- iv) Fixed rakes
- v) Aligning roller

i) FIXED GRID

Grid length	: 66 m
Notch pitch	: 66 mm
Grid material	: cast iron

The fixed grid is a simple cast iron construction with notches profiled identical to rake notches. They receive bars from diverter aprons. The moving rakes picks up the material from the fixed grid. Recess is provided for the movement of rakes. The fixed grid is provided in suitably sized segments.

ii) MOVING RAKES

Rake Length	: To fit size
Rake Pitch	: 600 mm
Notch Pitch	: 66 mm
Rake Material	: MS flats
Rake Mounting	: On MS fabricated frame
Rake Material	: MS flats
Rake Mounting	: On MS fabricated frame
Rake Thickness	: 16 mm

The moving rakes transports the bars towards the discharge end moving the material by one notch for every cycle, during the period cooling the bar by natural currents. The movement of the rakes is through the eccentrics.



iii) ECCENTRICS & DRIVES

Rake Drive : Through eccentrics

Eccentricity : 36 mm

Line Shaft Bearings : Spherical roller bearings housed in standard Plummer Blocks

Line Shaft Material: EN8

Eccentric Material : EN9

Line Shaft Inter Couplings: Muff type

Couplings between Line shaft and Worm Reduction Gearbox: Half Gear Coupling No. 108

Couplings between Worm Reduction Gearboxes & Reduction Gearbox and Motor: Half Gear Coupling No. 105

Drive mechanical : Worm Gear Boxes

The eccentrics are mounted on line shafts, housed in bearing housing bolted to moving rake mounting frame. The line shafts are coupled to gearboxes driven by a DC motor. The arrangement provides for one rotation of eccentric per cycle, which in effect moves the material by bone rake pitch.

iv) FIXED RAKES

Rake Length : To fit size

Rake Pitch : 300 mm

Notch Pitch : 66 mm

Rake Material : MS flats

Rake Mounting : On beams anchored to foundation.

The fixed rakes hold the bars during the rest of the periods.

v) ALIGNMENT ROLLS

Roll Diameter : ~200 mm

Bearings : Anti friction

Roll Lengths : ~ 600 mm

Notch Profile : To suit rake

Roll Drive : Geared motor

Towards the discharge end are placed a series of aligning rolls which aligns the bar front ends to facilitate cutting at cold shear. These are cast iron driven rolls profiled identical to the rakes and positioned between rakes.



c. Run Out Roller Table

1 Set

Run- out roller table

- a) Roller Pitch – 1500mm
- b) Roller size 216 Ø x 500 B.L.
- c) Spherical roller bearing
- d) MS Fabrication Table Frame
- e) Length 72 Mtr.

d. Cold Shear

1 No.

- Capacity : 300 Ton
- Blade width : 800 mm
- Nos. of cuts : 8 cuts/ min
- Construction : Fabricated steel body, heat-treated steel gears,
shafts & blades, shafts mounted on roller bearings
- Drives : AC Motor



9 Continuous operating dividing shear for TMT Bars

Type	: Continuous Shear
Bar Size	: 8 to 12mm
Bar Temperature	: 800°C
Maximum Bar Speed	: 12 m/s -22 m/s
Construction	: Steel fabricated body, heat treated steel gears and shafts, Antifriction Roller Bearings.
Lubrication	: Forced oil lubrication from Central Oil Lubrication System # 3.

☐ **Shear will be provided with:**

- Lubricating Piping on machine suitable for connection to central oil supply system.
- Entry & Exit Guides
- Driving D.C. Motor
- Bed Plates for Drive D.C. Motors in fabricated steel construction.

☐ **Materials of Construction:**

Body of Shear : Steel Fabricated & Stress Relieved

Gears & Pinion : 18CrNiMo7-6 case hardened & heat treated to 58–62 HRC Profile Ground
Bearings : Antifriction SKF or FAG make

Shafts : Heat Treated Steel

Blades : Heat Treated Steel (46-50 HRC)



10. Bar Finishing, Bundling & Tying Machine (Packing Bed)**10.1 Cold Shear Run – Out Roller Table (Gauge Conveyor)**

Purpose : Convey the material to bar transfer run-in roller table

Total Length : 26000mm

Table Length (C/C to end rollers) : 24000mm

Roller Spacing : 1200mm

Rollers : Diameter 190mm,

Barrel Length – 800mm,

Speed : 1.9 m/sec

Frame : Steel Fabricated

Drive : Group Driven with chain & sprocket through reduction gear box

Roller Neck Bearings : Pillow Blocks

Disappearing Stopper : 2 no. of pneumatically operated disappearing stoppers shall be installed at the exit of cold shear run-out roller table to stop the bar at 6/12 m from cold shear

10.2 Packing Bed

1) Size: 12 MW x 7.0 ML

2) Transfer Inlet-Arm unit x 12 M:

A) Main Shaft: Ø75 x 12 ML

B) Main drive motor: AC-10 HP x 4P x 1/10

C) Main Wrom Gear Reducer: A-200 x 1/30

D) Gear coupling: TY –SB- 150 x 2

3) Conveyor chain

4) Sprocket

5) Conveyor chain driven unit x 12ML

A) Main Shaft: Ø65 x 12ML

B) Main drive motor: AC –10HP x 4P x 1/10

C) Main Wrom Gear Reducer

D) Gear Coupling

10.3 Roller Conveyor

1) Total Length = 19,000 (mm)

2)V-roller before bundling machine = 5,000 (mm)

2-1) Conveyor drive unit:

i. Drive gear motor: AC-7.5 HP x 4P x 1/19

ii. Conveyor roller: Ø216.3 x 450W

iii. Coupling: RF-265

iv. Roller Chain & Sprocket: RS 100

2-2) Free V-roller unit:

a) V-roller: Ø 89.1 x 406W

b) Bearing:6207Z



2-3) Bundle are drive unit:

- a) Driver gear motor: AC-5HP x 4P x 1/19
- b) Conveyor roller: Ø216.3 x 450W
- c) Roller Chain & Sprocket: RS100

2-4) Clamp by air cylinder: TC- Ø250 x 400 ST

3) Roller Conveyor after bundling -= 14,000 (mm)

3-1) Conveyor drive unit:

- v. Driver gear motor: AC-7.5 HP x 4P
- vi. Conveyor roller: Ø267.4 x 600W

3-2) Clamp by air cylinder: TC- Ø250 x 400 ST

3-3) Stopper: TC- Ø250 x 530ST

10.4 Pneumatic Tying Machine - 2 Nos.



11. TMT System**TECHNICAL PARAMETERS**

- | | |
|---------------------------------|--|
| 1. Mill | Bar Mill |
| 2. Input Size | Billets 100 mm |
| 3. Sizes to be processed | 8 - 32 mm. |
| 4. Temperature after last stand | Max. 1000°C for 8 - 12mm dia.
Max. 1050°C for 16 - 25mm dia. |
| 5. Temperature difference | Less than 50°C between head and tail ends
after last stand |
| 6. Chemical Composition | As per IS 1786 : 2008 |
| 7. Aim | Production of High strength rebars as per
IS 1786 : 2008
Grades Fe 415, Fe 415D, Fe 500, Fe 550D,
Fe 550, Fe 550D and Fe 600. |

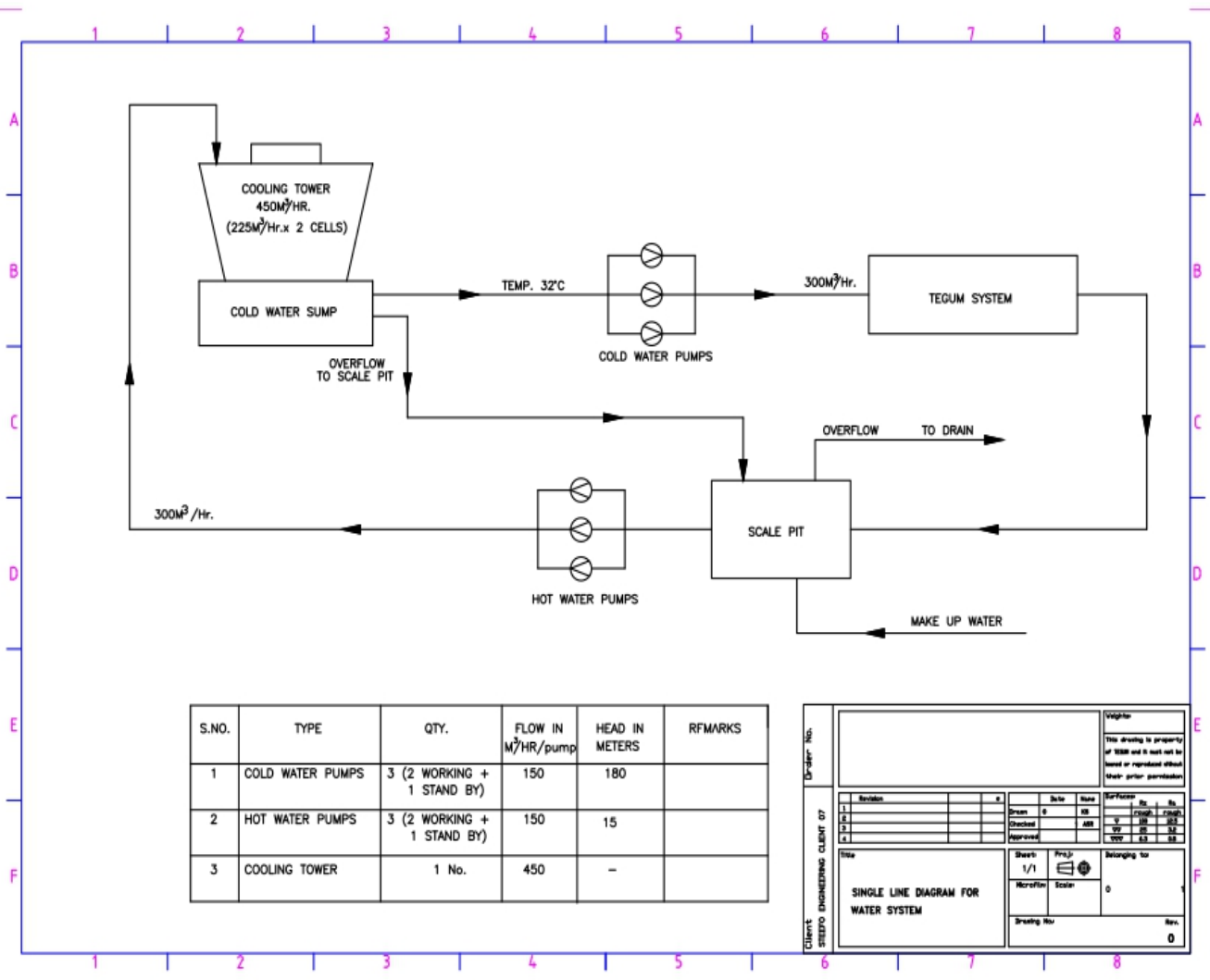


The diagram illustrates the TMT (Thermal Management System) components and their interconnections. Key elements include:

- Pressure Lines:** 60 N.B. AIR PRESSURE LINE at the top.
- Flow Indicators:** Arrows showing flow direction, such as "To Pyrometer" and "COOLING TOWER BASIN".
- Valves and Pumps:** Various symbols for control valves, pumps, and isolation valves.
- Legend:** A comprehensive key for symbols used in the diagram.

SYMBOL	DESCRIPTION
1-2	PIPELINE
3	PUMP
4	VALVE
5	ISOLATION VALVE
6	ISOLATION VALVE
7	ISOLATION VALVE
8	ISOLATION VALVE
9	ISOLATION VALVE
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11	ISOLATION VALVE
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95	ISOLATION VALVE
96	ISOLATION VALVE
97	ISOLATION VALVE
98	ISOLATION VALVE
99	ISOLATION VALVE
100	ISOLATION VALVE

Water Diagram



TECHNICAL DESCRIPTION

1 Introduction

For the purpose of controlled processing of deformed steel bars by means of high pressure water, Rolling Mill TMT System will be installed in the Rolling Mill with the aim to produce 'Quenched and Self-Tempered' (QST) steel rebars. A suitable shear needs to be located after the Rolling Mill TMT System to cut the rebars to suit the Cooling Bed length.

2. Technical parameters

The bars produced through the Rolling Mill TMT System as per this offer shall produce rebars as per the IS 1786: 2008 (Fourth Revision) amendment 3 of March 2017 Grades Fe 415, Fe 415D, Fe 500, Fe 500D, Fe 550, Fe 550D and Fe 600.

3. Layout

3.1 Main Rolling Mill TMT system with Pinch Roll Unit.

The main TMT system consists of the Rolling Mill TMT Pipes installed after the finishing rolling stand but before the cooling bed. Two pinch roll units need to be located before and after the Rolling Mill TMT pipes.

3.1.1 Arrangement of Rolling Mill TMT pipes

For dimension range of Ø 8 to 25 mm, adequate number and type of Rolling Mill TMT pipes of different lengths will be used. The length of the Rolling Mill TMT Pipes would suit mill parameters.

Fabricated base frame trolley(s) for mounting of the desired Rolling Mill TMT pipes will be provided for the system.

In the TMT Line, Pinch roll units with speed control are needed - to ensure that the tail end of rebar is thermo-processed in the same way as the front and middle portion of the rebar to obtain uniform properties over the entire length.

3.1.2 Water

High pressure water pumps will be provided so that water at desired pressure is available for rapid water quenching of rebars to obtaining desired properties.

3.1.3 Compressed air consumption

- Air Compressor

About 250 CFM at 7 bar will be needed to be provided by client.

3.1.4 Pressure water pump station

The pump station has high-pressure water pumps, including one standby pump, which will deliver water into a main pipe.

A motor operated valve controls water flow.



3.1.5 Details of the water and air-piping scheme

The Rolling Mill TMT pipes are fed from the pressure water manifold through flexible hoses that are controlled by manually operated hand valves. Each rolling mill TMT pipe has two water inlets.

The valves have only two positions "closed" and "open". This makes the operation very simple for the operators.

At the opposite end of the water inlets is arranged the air inlet. A manually operated valve controls the airflow.

3.1.6 Measuring and control instrumentation**3.1.6.1 Control instruments**

Located in a control cabin.

3.1.6.2 Pressure transmitter

A pressure transmitter with indicator and 4 - 20 mA signal mounted on water header near TMT system. The water pressure would be displayed on the digital meter located in the control cabin.

3.1.6.3 Temperatures

The final rolling temperature TF (measuring range 300 - 1300°C) is measured ahead of the TMT pipe.

The equalizing temperature TE (measuring range 300 - 1300°C) is measured on the cooling bed. For quality control purposes, the temperatures are indicated in the control cabin.

3.1.6.4 Check of tensile properties

At the beginning of a rolling campaign the tensile properties should be checked on samples as soon as possible and related to the equalizing temperature TE. This test is made by means of a tensile testing machine.

3.2 Cooling Tower & Scale Pit

The water used for water quenching as per the TMT Technology is collected in a scale pit via water flumes. It is then cooled at a Cooling Tower for re-use in the Rolling mill TMT system.

The hot water pumps and the Cooling Tower are included from our Scope of Supplies. Additionally, we shall provide the detail engineering for the water flumes and scale pit to enable the customer to build them as per local standards and practices.

3.3 Electrics & Controls

The electrics will operate from the supply voltage of 415V AC $\pm$ 10%, 50 Hz $\pm$ 5%.



SCOPE OF SUPPLY

The quantity and technical data given in this Chapter is indicative and subject to change. These will be finalized during detail engineering to ensure that the client is able to produce TMT rebars of international standard.

1 Main Rolling Mill TMT system

- 1 lot Rolling Mill TMT pipes with appropriate number of intermediate lengths.
- 1 lot manual valves PN 21 for water
- 1 lot flexible hoses PN 16 for water
- 1 lot manual valves R 1" PN 6 for compressed air
- 1 lot flexible hoses DN 20 PN 6 for compressed air
- 1 lot analogue pressure gauge display 0-25 Kg/cm² for water
- 1 lot motor controlled valve PN 21

2 Support Trolley for Rolling Mill TMT Pipes

The support base frame before cooling bed is to accommodate the different Rolling Mill TMT pipe lengths with holes bored to suit for mounting. When using short pipes, the remaining will be bridged with cast iron funnels

3 Piping

The interconnected pipes, with flanges and fittings will be supplied by us.

4 Temperature measuring instruments

- 1 No. Optical pyrometer, range 300 - 1300°C located immediately after the finishing stand;
- 1 No. Optical pyrometer, range 300 - 1300°C located in the centre of the cooling bed;

5 Pressure Transmitter

- 1 No. Pressure transmitter range 0 - 35 Kg/cm² located on the water header near TMT System.

6 Electrical equipment

The electrical equipment will be suitable to operate from 415 VAC $\pm$ 10%, 50 Hz $\pm$ 5% at 40°C ambient.

7 Controls

Controls would be located in the control room and pulpit and would be suitable to operate at an ambient temperature of 30°C.

8 High Pressure Pumps

Lot of high pressure pumps (including 1 standby) for pumping water from Cooling Tower basin to Rolling Mill TMT System will be supplied by us.

9 Cooling Tower

A Cooling Tower of adequate rating will be supplied by us.

10 Hot Water Pumps

Lot of hot water pumps (including 1 standby) for pumping water from the scale pit to the Cooling Tower will be supplied by us.



12. CENTRAL LUBRICATION OIL SYSTEMS

1. Four (4) Nos. Central Oil Lubrication System will be installed for forced feed lubrication of reduction gearboxes of the stands & shears

System #1 : To serve the lubrication requirement of bearings and gears of:
- Roughing Mill Gear Boxes

System # 2
Function : To serve the lubrication requirement of bearings and gears of:
- Gear Boxes Mill Stands C1 – C6
- Crop Cum Cobble Shear #1

System # 3
Function : To serve the lubrication requirement of bearings and gears of:
- Gear Boxes Mill Stands C7 – C14
- Crop Cum Cobble Shear #2

System # 4
Function : To serve the lubrication requirement of bearings and gears of:
- High Speed Pinch Rolls
- Slow Speed Pinch Rolls
- Dividing Shears
- Tail brake Pinch Rolls

System # 5
Function : To serve the lubrication requirement of bearings and gears of:
- Continuous Shear

Each system completes with:

- Oil tank complete with electrical heaters, level gauges, ventilator, inlet, outlet & drain connections. Tank capacity about 20 minutes of system flow.
- Working & standby motorized screw type pumps.
- Duplex Filters of suitable micron elements
- Plate type Heat Exchanger.
- Protection equipment & indicators such as Thermostats, pressure gauges, pressure switches, temperature indicators & switches, set of alarms & securities.



2. Air-oil Lubrication System One (1) No.

Function : One (1) for Roll neck bearings and roller guides of Continuous mill stands.

Technical Features:

Air Oil lubrication complete with:

- Oil tank with level gauges & heaters
- Pumps and motors complete with non return valves, unloading valves and suction filter.
- Delivery Filters
- Protection equipment and indicators such as thermostats, manometers, gauges and pressure switch, oil level indicator and alarms etc.



							S	STEEFO		
							C	CUSTOMER		
SCOPE OF SUPPLY OF AUTOMATIC ROLLING MILL										
Sr. No.	Description	Qty.	Basic Data	Basic Design	Detailed Design	Supply	Supervision of Erection	Erection & Comm.	Supervision of Comm.	
A	MILL EQUIPMENT									
1	550 MM Heavy Duty Roughing Mill									
1.1	High Speed Flywheel	1 No.	S	S	S	S	S	C	S	
1.2	Reduction Gear Box	2 Nos.	S	S	S	S	S	C	S	
1.3	550mm 3 -Hi Pinion Gear Box	1 No.	S	S	S	S	S	C	S	
1.4	Base Plate for Pinion Gear Box	2 Nos.	S	S	S	S	S	C	S	
1.5	550mm Mill Stand	1 No.	S	S	S	S	S	C	S	
1.6	Cardan Shaft with Universal Head	3 Set	S	S	S	S	S	C	S	
1.7	Full Geared Heavy Duty Couplings	4 Set	S	S	S	S	S	C	S	
1.8	Foundation Bolts	1 Set	S	S	S	S	S	C	S	
1.9	Motor Base Plate	3 Nos.	S	S	S	S	S	C	S	
2	400mm Continuous Intermediate Mill									
2.1	Reduction cum Pinion Gear Box	2 Nos.	S	S	S	S	S	C	S	
2.2	400 mm Mill Stands	2 Nos.	S	S	S	S	S	C	S	
2.3	Cardan Shafts	4 Nos.	S	S	S	S	S	C	S	
2.4	Geared Coupling	2 Nos.	S	S	S	S	S	C	S	
2.5	Foundation Bolts	1 Set	S	S	S	S	S	C	S	
2.6	Motor Base Plate	8 Nos.	S	S	S	S	S	C	S	
			S	S	S	S	S	C	S	
3	360mm Continuous Intermediate Mill									
3.1	Reduction cum Pinion Gear Box	4 Nos.	S	S	S	S	S	C	S	
3.2	360 mm Mill Stands	4 Nos.	S	S	S	S	S	C	S	
3.3	Cardan Shafts	8 Nos.	S	S	S	S	S	C	S	
3.4	Geared Coupling	4 Nos.	S	S	S	S	S	C	S	
3.5	Foundation Bolts	1 Set	S	S	S	S	S	C	S	
3.6	Motor Base Plate	16 Nos.	S	S	S	S	S	C	S	
4	320mm Continuous Intermediate Mill									
4.1	Reduction cum Pinion Gear Box	4 Nos.	S	S	S	S	S	C	S	
4.2	320 mm Mill Stands	4 Nos.	S	S	S	S	S	C	S	
4.3	Cardan Shafts	8 Nos.	S	S	S	S	S	C	S	
4.4	Geared Coupling	4 Nos.	S	S	S	S	S	C	S	
4.5	Foundation Bolts	1 Set	S	S	S	S	S	C	S	
4.6	Motor Base Plate	16 Nos.	S	S	S	S	S	C	S	



4	320mm Continuous Intermediate Mill								
4.1	Reduction cum Pinion Gear Box	4 Nos.	S	S	S	S	S	C	S
4.2	320 mm Mill Stands	4 Nos.	S	S	S	S	S	C	S
4.3	Cardan Shafts	8 Nos.	S	S	S	S	S	C	S
4.4	Geared Coupling	4 Nos.	S	S	S	S	S	C	S
4.5	Foundation Bolts	1 Set	S	S	S	S	S	C	S
4.6	Motor Base Plate	16 Nos.	S	S	S	S	S	C	S
5	280mm Continuous Finishing Mill.								
5.1	Reduction cum Pinion Gear Box	4 Nos.	S	S	S	S	S	C	S
5.2	280 mm Mill Stands	4 Nos.	S	S	S	S	S	C	S
5.3	Cardan Shafts	8 Nos.	S	S	S	S	S	C	S
5.4	Geared Coupling	4 Nos.	S	S	S	S	S	C	S
5.5	Foundation Bolts	1 Set	S	S	S	S	S	C	S
5.6	Motor Base Plate	16 Nos.	S	S	S	S	S	C	S
6	Auxiliary Equipment								
6.1	Y Table for Roughing Mill to feed Top passes automatic	1 No.	S	S	S	S	S	C	S
6.2	Fix Roller Table for Roughing Mill to feed Bottom passes automatic	1 No.	S	S	S	S	S	C	S
6.3	Approach Roller Table to 1st Continuous Mill (Extension Roller Table)	1 No.	S	S	S	S	S	C	S
6.4	Pinch roll Before Quenching Box	1 No.	S	S	S	S	S	C	S
6.5	Pinch roll after Quenching Box	1 No.	S	S	S	S	S	C	S
6.6	Start/ Stop type Crop & Cobble Shear	2 Nos.	S	S	S	S	S	C	S
6.7	Tail Breaking Pinch Rolls	2 Nos.	S	S	S	S	S	C	S
6.8	Roller Guide Box with Holder	1 Set	S	S	S	S	S	C	S
6.9	Roller Twist Guide	1 Set	S	S	S	S	S	C	S
6.10	Funnels	1 Set	S	S	S	S	S	C	S
6.11	Static Guides	1 Set	S	S	S	S	S	C	S
6.12	Water pump	2 Nos.	S	S	S	S	S	C	S
7	ELECTRICALS								
7.1	AC Motor for 550mm Roughing Mill	1 No.	S	S	S	S	S	C	S
7.2	D.C. Motor for 400mm Continuous Mill	2 Nos.	S	S	S	S	S	C	S
7.3	D.C. Motor for 360mm Continuous Mill	4 Nos.	S	S	S	S	S	C	S
7.4	D.C. Motor for 320mm Continuous Mill	4 Nos.	S	S	S	S	S	C	S
7.5	D.C. Motor for 280mm Continuous Mill	4 Nos.	S	S	S	S	S	C	S
7.6	Small AC Motors for Y Table, Turning Wall Roller table, Approach Roller table, Furnace Discharge Roller Table, etc.	1 Set	S	S	S	S	S	C	S
7.7	Main LT A.C. Incomer Panel for all AC / DC Motors	1 Set	S	S	S	S	S	C	S



7.8	D.C Drive Control Panel with Control Desk for Shears & Pinch Rolls	7 Nos.	S	S	S	S	S	C	S
7.9	Scada base DC Drives Control Panel with Control Desk for 400mm, 360mm, 320mm & 280mm Continuous Mill	14 Nos.	S	S	S	S	S	C	S
7.10	PLC, PCs, Remote I/Os etc System for Rolling Mill	1 Set	S	S	S	S	S	C	S
7.11	Motorised Liquid Resistance starter for 1120 KW Motor	1 No.	S	S	S	S	S	C	S
7.12	DC Motors for Shears & Pinch Roll	7 Nos.	S	S	S	S	S	C	S
7.13	ACB for DC Incomer	1 No.	S	S	S	S	S	C	S
7.14	APFC Panel	1 No.	S	S	S	S	S	C	S
8	Automatic Cooling Bed with Twin Channel & Cold Shear -66 Mtr.	1 Set	S	S	S	S	S	C	S
8.1	Twin Channel	1 Set	S	S	S	S	S	C	S
8.2	Rack type Automatic Cooling Bed	1 Set	S	S	S	S	S	C	S
8.3	Run Out Roller Table	1 Set	S	S	S	S	S	C	S
8.4	Cold Shear	1 No.	S	S	S	S	S	C	S
9	Continuous operating dividing shear for TMT Bars	1 No.	S	S	S	S	S	C	S
10	Bar Finishing, Bundling & Tying Machine (Packing Bed)	1 No.	S	S	S	S	S	C	S
10.1	Cold Shear Run – Out Roller Table (Gauge Conveyor)	1 No.	S	S	S	S	S	C	S
10.2	Packing Bed	1 No.	S	S	S	S	S	C	S
10.3	Roller Conveyor	1 No.	S	S	S	S	S	C	S
10.4	Semi Automatic Tying Machine	2 Nos.	S	S	S	S	S	C	S
11	TMT System	1 No.	S	S	S	S	S	C	S
11.1	Lot of Rolling Mill Thermex Pipes	1 Lot	S	S	S	S	S	C	S
11.2	Lot of Carriage & Pipe Fittings	1 Lot	S	S	S	S	S	C	S
11.3	Lot of Valves & fittings for Carriage	1 Lot	S	S	S	S	S	C	S
11.4	Lot of High Pressure Hose	1 Lot	S	S	S	S	S	C	S
11.5	Motor Control Valve	1 Lot	S	S	S	S	S	C	S
11.6	Pyrometers	1 Lot	S	S	S	S	S	C	S
11.7	Indicating Pressure Transmitter	1 Lot	S	S	S	S	S	C	S
11.8	Display and control system	1 Lot	S	S	S	S	S	C	S
11.9	Cooling Tower	1 Lot	S	S	S	S	S	C	S
11.10	High Pressure/Hot Water Pumps	1 Lot	S	S	S	S	S	C	S
12	Central Oil Lubrication System	6 Nos.	S	S	S	S	S	C	S



B. Wire Rod Block Mill – 6 Stand**WRM Design Basis**

Proposed equipment layout is indicated in drawing. Product which can be rolled through this line will be wire rods from dia **5.5mm to dia 8mm** in coil form.

Process

- Rolling stock from the last stand of the continuous mills will be directed to a **6 stand** “Twist free” block mill for producing finished Wire Rod
- At the entry, a horizontal pinch roll will be located which will assist the bar in travelling to block mill.
- Downstream, a disc & drum shear is located to crop the nose & tail of bar before entry to block. This shear will also cut the bar into fragments in the event of a cobble occurring downstream in the mill.
- After the shear, a side looper will be located to create a loop before the block mill. This loop is required to maintain the stock size at the block mill. Scanner located on this looper will be used to control the bar mill rolling speed to control the loop. Block mill speed will be fixed and speed of mill upstream will be varied.
- The twist free finishing block is of an advanced design, engineered to meet the strenuous demands of today’s high speed rod mills. The configuration comprises an electric motor driving through a synchronizing gearbox to give twin output extended to bevel gearboxes to from the two main mill drive-shafts of the block. From these drive shafts the bevel gears take the drive to the stands at 45o on each side of the vertical centerline so giving 90o stand to stand relationships. The driveshaft’s and bevel boxes etc, is all below pass line thus giving a low overall mill height for maximum operator access, visibility and operation. The mill is designed and built on a modular design and carry all rolls and rolls holders of identical dimensions and all entry and delivery guide holder are identical on all stands.

The roll / roll holders are mounted onto the roll shafts by means of hand operated hydraulic mounting tools and are dismounted using similar units. This allows rapid roll changing giving positive benefits in production time. The majority of stands having two pass grooves on the rolls and as the roll / roll holders are reversible, a further saving in carbide utilization is achieved.

- The stiffness of the stand, enabling it to roll closed tolerance, is gained from design features limiting the rolling loads stress path to a minimum and by pre-stressing the unit by means of a hydraulic cylinder. The design and cylinder facility also allows infinite roll center setting and adjustment within the maximum to minimum roll diameter range of 216-205 nominal. Axial adjustment is also provided on one roll shaft to allow accurate alignment of roll grooves.



- Finished stock from block mill will pass through a water cooling unit to reach the horizontal coil laying head. Flow of water through this unit is controlled for rapid cooling to drop the laying temperature of rolled plain bars to desired level.
- Plain wire rod laid in a spiral form by horizontal coil laying head on the air cooling conveyor will reach the coil reform station and gets formed in to a coil on the mandrel of a retractable type coil mandrel located below reform tub.
- Motor driven roller table will transfer the loosely formed coil on to the coil mandrel to a on line coil compactor before shifting to a 4 arm coil capstan.
Compacted coils from coil capstan need to be shifted to storage bay either by an overhead crane or by fork lifts.

SAMPLE ROLLING PARAMETERS

No	Bar size mm	Max. Bar speed m/s	Production Tons/hr.	No. of Stands
1	Dia. 5.5	60	30	6
2	Dia. 6	50	30	6
3	Dia. 8	30	30	2

GRADE OF STEEL ROLLED

Low Carbon steels.



CONTROLLED COOLING

Finished wire rod from Twist free block mill and laid on the roller conveyor in spiral layers by Coil laying head will be cooled in a controlled manner on the roller conveyor located between coil laying head and coil reforming station.

Cooling of wire rod will be in following ways depending on requirement of grade of steel;

- **Natural cooling:** Wire rod will be cooled with natural air between coil laying head and coil reforming station
- **Fast cooling :** Air blowers located below the conveyors will force the air to cool the wire rod. Cooling rate will be controlled by controlling the air quantity / using limited number of blowers
- **Retarded cooling :** Retractable Insulated hoods mounted on the roller conveyor frames will cover the conveyor upper side to retard the natural cooling rate.

No	Type of cooling	Grades of steel
1	Fast Cooling qualities	Typical S.W.P, Tyre bead, plain carbon of 0.3 to / 0.85 % C content
2	Slow Cooling qualities Typical	E.W.R, SAE 8620, EN45, 65Si7, 20Mn Cr5,/ S.A.E 52100, 16 MnCr5, 0.23%C



2. Equipment Technical Details**2.1 Pinch Roll**

Pinch roll unit is installed to assist the rolled bar travel from intermediate stand to the block mill.

The equipment consists of a set of rolls made of wear resistant casting, mounted on shafts supported on antifriction bearings. Top and bottom roll bearing housings are located on fabricated links pivoted on pivot frame. Top roll link is connected to a pneumatic cylinder which will control the pinching pressure of rolls on the bar.

Drive to the rolls from Motor is transmitted through a pinion gear box with hardened and ground gears. Gears and bearings of the gear box are splash oil lubricated. Pinch roll unit, gear box and motors are mounted on a common base frame.

Type	Cantilever type
Roll dia	250 mm Nominal
Roll material	Wear resistant cast iron
Drive Method	By electric motor through pinion gear box & propeller shafts
Pinching action	Pivoted top roll actuated by a pneumatic cylinder
Motor	30kW, 1500 RPM DC
Motor Control	DC variable speed drive
Speed	25 m/sec max. & 10 m/sec min.
Bar Size	Dia. 8mm to Dia.25mm
Gear Lubrication	Splashed oil lubrication

2.2 Crop & Cobble Shear

Type	Disc & Drum Shear
Max. bar Dia	13mm
Max. bar Speed	22 m/sec
Shear Blade Centres	300mm
Motor capacity	11 kW, AC / 15 KW AC
Blade Hardness	50 (+/- 2) HRC
Blade Material	Ni-Cr-Mo Steel

2.3 Side Looper

Type	Horizontal
Swing Arm	Pneumatic Operated
Max. Bar Speed	23 m/sec



2.4 6 Stand Twist Free Finishing Block Mill

The finishing Mill will comprise 170 / 155 mm cantilever type stands, arranged in a 45° opposed configuration, being alternately right and left hand.

Specifications:

No. of Stands	6
Angle of roll axes	45° V Type No-Twist
Max design speed	60 m/sec
Product size range	5.5mm to 8mm
Maximum Input size	11mm
Grades to be Rolled	Low Carbon Steel
TC Ring Size	OD 170mm max & 155mm min.
TC Ring Weight	11.7 kg approx.
Width & Number of pass per roll	70mm, 2 passes per ring
Roll Neck Dia	140mm
Roll balancing	Hydraulic cylinder accumulator @ 200kg/cm ²
Drive	DC Motor 750 kW 0-600-1200rpm
Lubrication System Capacity	700 lpm
Water flow Rate for Ring Cooling	200 m ³ /hr at pressure of 4-8kg/cm ²
Air Flow Rate	200 m ³ /hr
Average Reduction per pass	19%
Roll Stand Front Plate & Guides	Stainless Steel



Construction:**Bevel Gear Box**

- Drive to rolls will be transmitted through a bevel gearbox
- Bevel gearbox equipped with forged steel precision cut hardened and profile ground Spiral bevel gears
- The input and output shaft will run in cylindrical roller bearings and Angular contact ball bearings
- Gears and bearings will be arranged for oil lubrication from a Centralized oil system

Gears	Single helical dynamically balanced
Gears rating	400kW
Quality	En353 Case Hardened & Ground 60 +/- 2 HRC
Lubrication	Forced

Roll Stand

- Fabricated steel box housing of robust construction
- Forged steel single helical top and bottom roll shaft carried in cast steel gear links within the housing
- The roll shaft will run in white metal bearings at the roll end and roller bearing at the other, a double angular contact bearing being employed to accommodate axial loads
- Forged steel single helical drive shaft carried in the links will transmit the drive to the roll shafts
- The drive shafts will run in roller bearings with “four point” ball bearing taking the gear thrust loads
- The gear and bearing will be oil lubricated from a centralized oil system
- The roll neck labyrinth sealing will be supplemented by air / oil lubrication
- The roll gap will be manually adjusted by means of synchronized Twin screw and wedge arrangement acting on the gear links
- The links will be maintained in contact with the wedge gear by a hydraulic pre-stressing cylinder
- Axial roll adjustment gear for roll groove alignment will be In-corporated on the top roll shaft
- The adjustment gear will be manually operated

Size	216mm
Gears	Single helical dynamically balanced
Gears rating	400kW
Quality	En353 Case Hardened & Ground 60 +/- 2 HRC
Front Seal Plate & Covers	Stainless Steel
Gap Adjustment	Synchronised twin screw & wedge
Roll Balancing	Hydraulic cylinder at 200 kg/cm ²
TC Ring mounting	Held via taper sleeve pressed by hydraulic mounting unit
Axial Adjustment	+/- 2.0mm online without packing / removal of ring
Roll cooling	Stainless steel nozzles with flat / round spray



Synchronizing Gearbox

- The drive will be transmitted to the individual stand drive bevel gearbox through a common synchronizing gearbox
- Fabricated steel gearbox equipped with forged steel through hardened single helical gearing and twin output shafts
- Gears and bearings will be arranged for oil lubrication from a centralized oil system
- Steel cordon shafts equipped at each end with a flexible gear type
- Coupling will be provided from the synchronizing gearbox to the bevel gearbox, from bevel gearbox to bevel drive gearbox
- For each bevel drive gearbox, a double engagement flexible gear Type coupling will transmit the drive to the individual stands

Gears	Single helical dynamically balanced
Gears rating	4000kW
Quality	Case Hardened & Ground 60 +/- 2 HRC
Lubrication	Forced

Oil Lubrication System

Centralized oil lubrication system to serve the No-twist mill. System will comprise the following :

- One tank complete
- Two positive displacement pumps (1 running, 1 standby. Each pump will be complete
- with a bed-plate, motor coupling and coupling guard
- Suction strainers
- Direct acting relief valve for pump protection
- In line filters having manual changeover facility, giving 20 micron filtration
- Low level alarm in tank
- All necessary pipe-work, valves, switches, indicators and instrumentation at the pumping station only

Description	Block Lubrication
Pump Type	Screw/Gear Pump
No. of Pumps	1 + 1 standby
Flow & Pressure	600 lpm at 6 bar
Filtration Level	20 micron
Tank Capacity	6000 litres
Heat Exchanger	Plate Type

Centrifuge

One centrifuge will be provided for the purification of lubrication oils by removal of solids and water. Capacity is 400 lph and filtration level is 5 micron.

Emergency Pressure Vessel

An emergency pressure vessel of 1000 liters capacity will be provided to give approximately 2 minutes lubrication to the finishing mill in the event of a power failure.



ARRANGEMENT OF OIL LUBRICATION SYSTEM (TYPICAL ONLY)**Pre-Stressing Hydraulic System for Roll Balancing**

One (1) hydraulic system to serve the pre-stressing cylinder on the finishing block.

- Hydraulic medium : mineral oil
- Working pressure : 200 bar

System will comprise the following:

- One tank complete having a capacity of 150 liters
- Two positive displacement pumps (1 running, 1 standby) each having a delivery of 4.5 liters per minute at 200 bar. Each pump will be complete with a bell housing & motor coupling
- One relief valve for system overload.

Protection Cover Hydraulic System

One (1) hydraulic system to serve the cylinders for raising / lowering the hoods for the block mill stands. Hoods can be individual for each stand or common.

- Hydraulic medium : mineral oil
- Working pressure : 250 bar

System will comprise the following:

- One tank complete having a capacity of 150 liters
- Two positive displacement pumps (1 running, 1 standby) each having a delivery of 4.5 liters per minute at 10 bar. Each pump will be complete with a bell housing & motor coupling
- One relief valve for system overload.

Guides (Stainless Steel)

The followings guides will be provided for the finishing mill

- Roller entry guides with air oil lubrication for bearings.
- Exit stripper guides.

All Entry Roller Guides, Static Exit Guides, Guide Holders & Dummy Tubes will be made of Stainless Steel



Cobble Block

A cobble block will be provided immediately after the last stand finishing block to provide a 'break-out' point should a cobble occur downstream finishing line. Stainless steel cobble block with hinged, loose fitting top section will be located within the finishing block cobble guard.

Roll Holder

The roll holder will be mounted onto the roll shaft by means of tapered sleeves driven on by a hydraulic unit and removed by a similar dismounting unit

Roll Mounting & dismounting equipment

- The equipment will be designed for the mounting and dismounting of Roll and roll assemblies from the roll shaft of the stand.
- Roll assemblies will be secured to the roll shaft by tapered locking sleeves Hydraulically driven into the bores of the rolls and roll assemblies
- A specially adapted hydraulic unit will be provided for this purpose together with a similar unit for removal of the tapered sleeves

Portable & Bench Optic System

One portable & one bench optic will be provided for setting of guides on block mill & in guide shop

Roll Unit Changing

Since Roll unit & bevel units are independently enclosed and connected by gear coupling, separate tackle is not required. Hooks are provided on stands for lifting by crane. Roll unit can be opened separately in dust free room without risk of contamination on field

Air Oil Lubrication System

Air Oil lubrication system provides air oil mist for guides and also seals of roll stands to prevent water contamination. This improves life of seals also. Timing of this system is PLC Controlled

Temperature Indication of Bearings

Temperature of bearings will be displayed in Pulpit by connecting bearing RTDs to analog inputs of the main PLC. Alarm & Tripping will be done by system in case of overheating.



2.5 High speed Pinch Roll (1 no.)

Located before the coil laying head, the unit assists rod entry into the coil laying head.

Design

Diameter of rolls :	300mm
Rod Size :	5.5mm to 8mm Dia
Motor :	60 KW, 1500 RPM, AC

Technical Specifications

Fabricated steel box housing of robust construction. Forged steel single helical top and bottom roll shafts carried in fabricated steel links within the box housing. Forged steel single helical drive shaft carried in the links will transmit the drive to the roll shaft. Roll shafts and drive shafts carried in anti-friction bearings. Each link will be connected to a crank sleeve mounted within the housing. A pneumatic cylinder rotates both crank sleeves to pivot the link and bring the pinch roll together to provide the pinch.

Drive

Pinch roll unit driven by AC electric motor through a flexible gear type coupling. One drive shaft is extended through the housing to form the input shaft. Drive shaft will be meshed together to transmit the drive one to the other.

Rolls

Wear resistant graded cast iron rolls mounted on to the tapered roll-shafts

Pinch Mechanism

A pneumatic cylinder is directly connected to the crank sleeve and by an extension rod to the bottom crank sleeve. Adjustment to remove backlash from the system is provided within the extension rod. The roll travel to provide the pinch is governed by fixed length stroke of the pneumatic cylinder.

Adjustment of the position of the cylinder and therefore open dimension of the pinch rolls is used along with the fixed cylinder stroke to adjust the final roll gap when pinched. Cylinder position adjustment is by a ratchet mechanism external to the pinch roll housing.

Lubrication

The bearings and gears of the pinch roll unit is oil lubricated from lubrication system of block mill.



2.6 Coil Laying Head :**Function**

The coil laying head receives rod from the water boxes / pinch roll unit and lays it in an overlapping spiral configuration onto the coil conveyor

Design

Diameter	:	Nominally 1050
Balance	:	ISO 1940 – 1973 Grade 25
Angle of Inclination	:	15°
Rod Sizes	:	Dia. 5.5mm to Dia. 8mm
Motor	:	75 KW, 1500 RPM

The laying head consist of a rotating shaft to one end of which is attached a rotor which support the formed laying pipe. The rotating shaft is hollow and the rod is fed through the center of this shaft up to the laying pipe by a water cooled tubes.

Drive

The laying head is driven by a AC electric motor connected to the laying head input shaft by a flexible coupling. Drive is transmitted from the input shaft to the laying shaft by helical gearing.

Laying Pipe

The laying pipe is in single piece mounted on to coil cone with fabricated clamps.

Bed Plate

Bed-plate of fabricated steel to support the coil- laying head. The laying head motor is mounted directly onto the back of the laying head gearbox base frame.

Construction

The laying head casing and laying pipe guard is of fabricated steel. Laying shaft and rotor of alloy steel. Helical gearing of alloy steel. The input shaft and laying shaft are supported in anti-friction bearings.

Lubrication

The bearing and gear of the laying head are oil lubricated from lubrication system of block mill.



2.7 Laying Pipe Bending Jig

Function

For the manufacture of replacement laying pipes.

Design

The new pipe is located in the bottom of jig, heated locally, and bent around the pre-formed shape of the jig. The pipe is clamped to the as it is formed.

Construction

Fabricated steel bending jig with machined support positions. Steel clamps to hold the pipe in position during bending.



CONTROLLED COOLING EQUIPMENT**2.8 Coil Cooling Conveyor:**

Type	Roller Conveyor
Roller Conveyor length	30 meters
Roller Diameter	100mm (solid steel rolls for first segment of 4 meters) Hollow rollers there after
Roller length	1400mm
Roller Pitch	approx. 105mm at laying head and 200mm there after
Bearings	Pillow blocks
Number of drives	Five
Drive system	AC geared Motor , sprocket and chains
Drive motor	AC variable speed
Rating	11kW 625 rpm
Chain	1 inch pitch, single chain

Function

The coil-cooling conveyor will transport and advance the open laid coil of rod issuing from the coil laying head through the air-cooling section and upto the reform tub.

Design

- Overall conveyor length 65 meters
- Roller Conveyor
- Solid and pipe steel rollers carried in pillow blocks
- Fabricated steel conveyor frames supports bearings
- Rollers interconnected with chain and sprockets
- chain guard of fabricated steel
- Stub rollers at the step end of the conveyor
- Adjustable Section at the laying Head
- A section of conveyor at the coil laying head will be hinged and may be elevated or lowered to provide access for laying pipes removal, and to obtain optimum rod ring formation on the conveyor

Roller Section at Reform Tub

- Generally as the main roller conveyor
- Stub rollers at the reform tub
- Drive taken by transmission chain from the chain conveyor
- Adjustable longitudinally to provide optimum rod ring presentation into the reform tub
- Adjustment is manual

Adjustable Side plates

Manually adjustable side plates will be provided at the step between the roller conveyor and the chain conveyor and at the reform tub.



2.9 COIL FORMING AND HANDLING EQUIPMENT**2.9.1 REFORM STATION:**

One (1) fabricated steel reform tub mounted at the end of cooling conveyor on fabricated steel support structure

The coil plate will be driven by a hydraulic cylinder

Also incorporated will be hydraulic cylinder operated reform gate which will control the outside diameter of the coil

A fabricated steel pilot cone will be provided in the reform tub which will aid coiling of the rod whilst awaiting mandrel availability

The cone will rest loosely on the end of the mandrel when it is in a vertical position within the reform tub

The reform tub will receive coil spirals from the cooling conveyor and direct them on to the disappearing mandrel.

2.9.2 RING DISTRIBUTION SYSTEM (Optional)

Ring distribution system ensures distribution of rings to enable smaller coil height for given weight. Ring distribution system will be of either slewing bearing type operated by geared motor or variable protrusion type

2.9.3 COIL IRIS PLATE & CONE

4 nos of coil iris plate will be on either side of the cone. The plates will be driven by a hydraulic/ pneumatic cylinder. A fabricated steel pilot cone will be provided in the reform tub which will aid coiling of the wire rod whilst awaiting mandrel availability. The cone will rest loosely on the end of the mandrel when it is in a vertical position within the reform tub. Coil Plate opens and rests the cone on the mandrel

2.9.4 DISAPPEARING MANDREL WITH EASY DOWN

A disappearing mandrel surrounded by a cage will receive the coil on the roller conveyor. A vertically moving mandrel to support the Nose Cone for Coil Inside diameter control hydraulically operated. Along with the centring mandrel, a coil receiving mechanism called Easy Down mechanism supports the coil and slowly brings it down as the coil height builds up. This helps in maintaining a proper coil shape which is easy to compact, bind and transport. (The mandrel disappears / comes down when this easy down mechanism comes to its lowest level on the roller conveyor.) The easy down mechanism works hydraulically and is located inside the mandrel / boom. Since it goes through the roller conveyor, coil cage has its arms extended in only the axis which is parallel to the roller conveyor.

2.9.5 ROLLER CONVEYOR

Roller conveyor is located below the reform station. It will convey vertical coil collected in mandrel and passes it online coil compactor. Length is approx 6.5m. Roller conveyor is operated by geared motor.



2.9.6 ONLINE UP COMPACTOR

Online Up-compactor will lift and compress coil upto 1400kg. This operation is done by hydraulic cylinders. During this period, coil will be strapped manually before moving the coil ahead.

2.9.7 DELIVERY ROLLER TABLE

Delivery Roller Table is located below and after compactor. Delivery Roller Tables transfers compacted tied coils to Upender through geared motor.

2.9.8 WEIGHING STATION (Optional)

One delivery table is suspended on load cells for online coil weighing & printing for the tags

2.9.9 UP ENDER KICKOFF

Travelling type Up-ender lifts the compacted tied coil and kicks it off below or places it on the arms of the capstan.

2.9.10 COIL CAPSTAN (Optional)

The coil capstan has 4 arms to hold the compacted coils for further processing.



3.0 Scope of supply

S = STEEFO

C = CUTSTOMER

SCOPE OF SUPPLY OF WIRE ROD MILL

No.	Description	Qty.	Basic Data	Basic Design	Detailed Design	Supply	Supervision of Erection	Erection & Comm.	Supervision of Comm/
A	Entry Pinch Roll								
	Pinch Roll Mechanicals with motor base frame	1	S	S	S	S	S	C	S
B	Disc & Drum Shear								
	Disc & Drum Mechanicals with motor base frame	1	S	S	S	S	S	C	S
	Crop Bucket	1	S	S	S	S	S	C	S
C	Side Looper								
	Side Looper Mechanicals	1	S	S	S	S	S	C	S
	Loop Scanner	1	S	S	S	S	S	C	S
D	6 stand Block mill consisting of:								
	Synchro gearbox	1	S	S	S	S	S	C	S
	Gear couplings and floating shafts between gearboxes & stands (set)	1	S	S	S	S	S	C	S
	Base frame for synchro gearbox	1	S	S	S	S	S	C	S
	Base frame for driving motors	1	S	S	S	S	S	C	S
	Base frame for block mill & stand supports	1	S	S	S	S	S	C	S
	Pipeline & fittings on stand for oil, air & hydraulic systems (set)	1	S	S	S	S	S	C	S
	Standing Platform & Checkered plates around block mill (set)	1	S	S	S	S	S	C	S
	Snap shear at the entry	1	S	S	S	S	S	C	S
	Roll stands	6	S	S	S	S	S	C	S
	Stand supports	6	S	S	S	S	S	C	S
	Bevel gearboxes	6	S	S	S	S	S	C	S
	Cobble box at the exit	1	S	S	S	S	S	C	S
	Roll Holders complete assembly	12	S	S	S	S	S	C	S
	SS Entry guides	6	S	S	S	S	S	C	S
	SS Delivery guides	6	S	S	S	S	S	C	S
	Roll mounting tool	1	S	S	S	S	S	C	S
	Roll dismounting tool	1	S	S	S	S	S	C	S
	Hand operated hydraulic pump for above tools	1	S	S	S	S	S	C	S



E	Auxiliary equipment for block mill								
	Circulating Oil lubrication system for gear boxes & roll stands	1	S	S	S	S	S	C	S
	Hydraulic system for roll pre-stressing balancing	1	S	S	S	S	S	C	S
	Hydraulic system for roll guards operation	1	S	S	S	S	S	C	S
	Air oil lubrication system for guides & roll shaft seals	1	S	S	S	S	S	C	S
	Water separator for roll stand lubrication system	1	S	S	S	S	S	C	S
	Emergency Tank	1	S	S	S	S	S	C	S
	Gauge panels in lub pit & block mill	3	S	S	S	S	S	C	S
F	Equipment After Block Mill								
	High speed Pinch Rolls	1	S	S	S	S	S	C	S
	Horizontal Laying Head	1	S	S	S	S	S	C	S
	Pipe Bending Jig	1	S	S	S	S	S	C	S
G	Coil Cooling Conveyor								
	Solid Roller table mechanicals	4m	S	S	S	S	S	C	S
	Hollow Roller tables mechanicals	26m	S	S	S	S	S	C	S
	Under structure for Coil Cooling Conveyor	30m	S	S	S	S	S	C	S
H	Coil Handling System for Coil								
	Reform Station with coil gate	1	S	S	S	S	S	C	S
	Roller Table	1	S	S	S	S	S	C	S
	Coil Up-Ender	1	S	S	S	S	S	C	S
	4 Arm Coil Capstan	1	S	S	S	S	S	C	S
	Power pack for above equipment	1	S	S	S	S	S	C	S
I	Auxiliaries for Wire Rod Line								
	Connecting funnels between mill & WRM	1	S	S	S	S	S	C	S
	Foundation Bolts for all equipment (set)	1	S	S	S	S	S	C	S
J	Main Motors								
	DC Motor for Pinch Roll	1	S	S	S	S	S	C	S
	AC Motors for Disc & Drum Shear	2	S	S	S	S	S	C	S
	DC Motors for Block Mill	1	S	S	S	S	S	C	S
	DC Motors for High Speed Pinch Rolls	1	S	S	S	S	S	C	S
	DC Motor for laying head	1	S	S	S	S	S	C	S
	AC Geared motors for coil cooling conveyor	4	S	S	S	S	S	C	S
	AC Geared motors for coil handling system	3	S	S	S	S	S	C	S
K	Automation & Controls								
	LT Panels with drives	1 Set	S	S	S	S	S	C	S
	Field Instrumentation - 4 nos. HMD, 2 nos. Pyrometers, 1 no Loop Scanners, Temperature & pressure transmitters	1 Set	S	S	S	S	S	C	S
	Proximity Switches	1 Set	S	S	S	S	S	C	S



PREFERRED MAKE OF STANDARD BOUGHTOUT ITEMS

AC Motors	: ABB / Kirloskar / Siemens / Bharat Bijlee
D. C. Motors	: Integrated Electric Co. / Kirloskar / Crompton Greaves
D. C. Panels for Continuous Mill Automation	: ABB / Parker
ACB, Switchgears, etc in AC Panels	: L&T / Siemens / ABB
Overload Relays	: Siemens / L&T / ABB
HRC Fuses	: L&T / ABB / Siemens
Starters	: Siemens/ ABB/Schneider / L&T
Push Buttons	: Siemens / L&T / Telemechanique / IFM
PLC	: Siemens
Control / Selector Switch	: Kaycee / Jyothi / Telemechanique
Bearings & Plummer Block	: SKF / FAG / TIMKEN
MCBs	: ABB / L&T / Siemens
Meters (Ammeter, Voltmeter, P.F. Meters):	AE / Rishab / IMP
Multifunction Meters	: Konzerv / Trinity
Indicating Lamps	: Siemens / Tekmic / Telemechanique
Pumps	: Kirloskar Pumps Ltd. / KSB Make
M. S. Plates, M. S. Channel, Structure, Pipes, etc	: As per Indian Standard IS – 2062 & other Grade from Primary producer such as Jindal Steel / Essar Steel / Steel Authority of India
Hydraulic Equipment	: Rexroth / Parker / Yuken
Air Compressor	: Atlas Copco
Lubrication System	: Shaan Lube Equipments
Cardan Shaft & Universal Cardan Shaft	: Nu Hydro / Cardan India
Chain & Sprockets	: Reynold / Dimond
Geared Coupling	: Nu Teck Couplings
TMT System	: Thermax

All material will be of Standard Make & equivalent to 1st Class Maker



C. Consultation & Supervision of Erection, Installation & Commissioning of Rolling Mill**A. Scope of Services.**

1. Preparing Plant Lay-out
2. Detail Engineering of Mechanical, Electrical and civil work involved in the project.
3. To provide technical assistance & supervision for successful commissioning of the projects.
4. We shall depute a team of suitable persons for erection and Installation of the Steel Plant.
5. Supply of foundation plans with load data for all the equipment including requirement of embedded parts, trenches for cabling & piping etc. to enable customer arrange for design and construction of civil works.

B. Facilities to implement project.

The client will provide following -

1. All local manpower skilled and unskilled workers required for erection.
2. All civil work will be done by the client as per our drg. & instruction of our Engineer.
3. All tools, tackles and material handling equipment like cranes etc. raw materials, consumables, Oil & Lubricants, Water, compressed air, power etc. will be provided by the buyer at buyer's cost to facilitate erection & commissioning.
4. All expenses Air Ticket, visa charges, insurance, lodging, Boarding, local conveyance etc. shall be borne by the client.
5. You will cover all our personnel engaged at project site by suitable insurance policy according to status of the person.
6. You have to pay Salary of all Persons coming on Site for Erection & Installation
7. You have to provide all Staff for Hot trial & Production of the Plant.



EXCLUSIONS

1. Shed (Hanger) & Crane
2. HT Side, Transformer, all Cables & Earthing
3. All Civil Work and Mill Foundations including water & electrical trenches.
4. Fabrication on Site like – Trough, Charging Rack, Walk Ways, etc.
5. Interconnecting Air, Oil, Water & Hydraulic Pipe Lines
6. Re Heating Furnace
7. Workshop Machinery including, Air Compressor CNC Rib Cutting & Roll Branding Machine
8. Material Testing Equipment
9. Bought out Items like Rolls, Bearings, etc.
10. Erection and Installation
11. First Fill of Lubricants

